Sexually Transmitted Infections and Pelvic Inflammatory Disease

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Conflicts of Interest

- I have no disclosures.

Learning Objectives

- Describe the major bacterial causes of STIs and their microbiologic characteristics.
- Differentiate the clinical presentations of gonorrhea and chlamydia and describe their major complications.
- Compare the stages of syphilis and describe their clinical manifestations.
- Differentiate the pathogens and clinical manifestations of the major causes of genital ulceration.
- Describe first-line treatments for gonorrhea, chlamydia, and syphilis and recall their mechanisms of action and major adverse effects.

Sexually Transmitted Infections (STIs)

- Definition:
  - infection that is passed from one person to another through sexual contact via vaginal, oral, and anal sex.
  - Some infections can be transmitted through skin-to-skin contact or via fomites.
  - Infections include viruses, bacteria, fungi or parasites

STIs

- These include, but are not limited to:
  - HIV
  - Syphilis caused by *Treponema pallidum*
  - Gonorrhea caused by *Neisseria gonorrheoae*
  - Chlamydia caused by *Chlamydia trachomatis*
  - Herpes Simplex Virus (HSV)
  - Human Papilloma Virus (HPV)
  - Trichomonasis caused by *Trichomonas vaginalis*
  - Other genital ulcerative diseases

Assessing Your Patient

- **History**
  - Set the stage
  - Obtain a sexual history in an **open, nonjudgemental** manner
  - Identify sexual orientation and gender identity



The following risk assessment questions are organized according to these categories.

Aberg JA, et al. Clin Infect Dis. 2014;58:e1-e34.

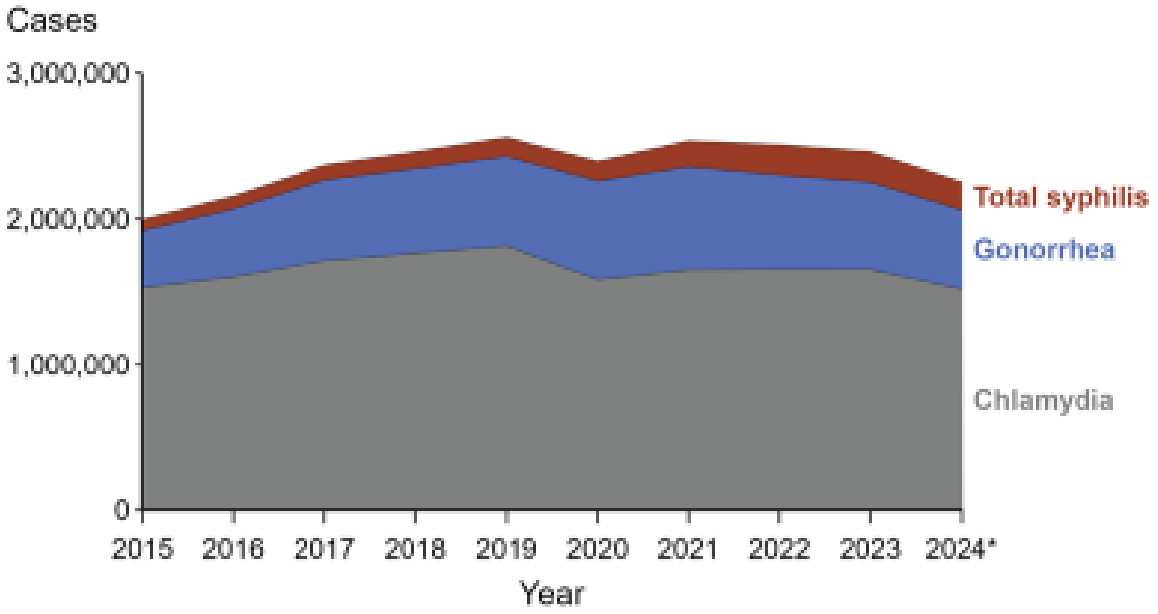
Assessing Your Patient

- **Physical**
  - overall exam
  - anogenital exam
  - gynecologic exam if appropriate
  - skin and mucous membranes
  - lymph nodes

Aberg JA, et al. Clin Infect Dis. 2014;58:e1-e34.

Chlamydia trachomatis

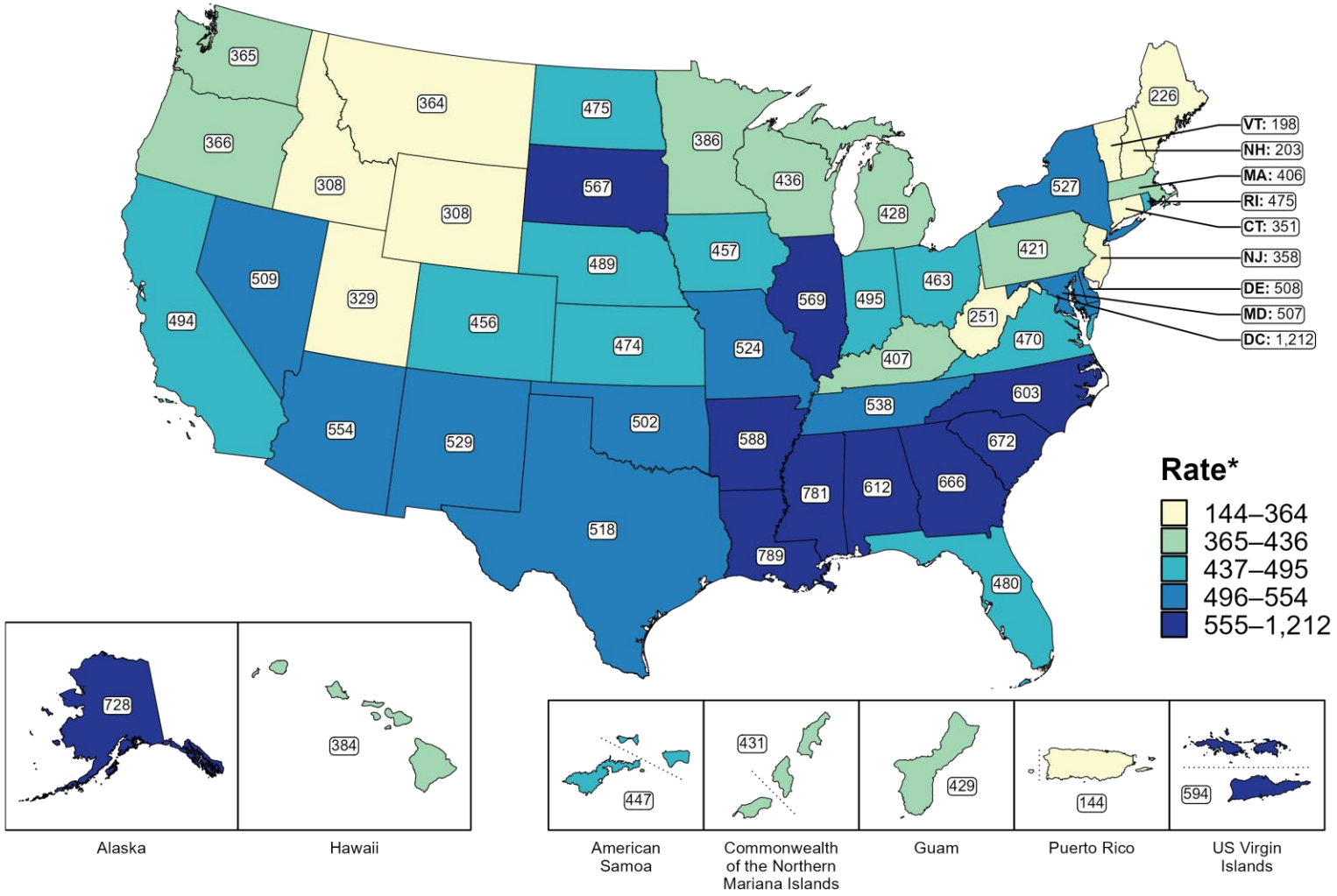
Sexually Transmitted Infections (STIs) — Reported Cases by STI and Year, United States, 2015–2024



* 2024 data are provisional as of August 14, 2025.

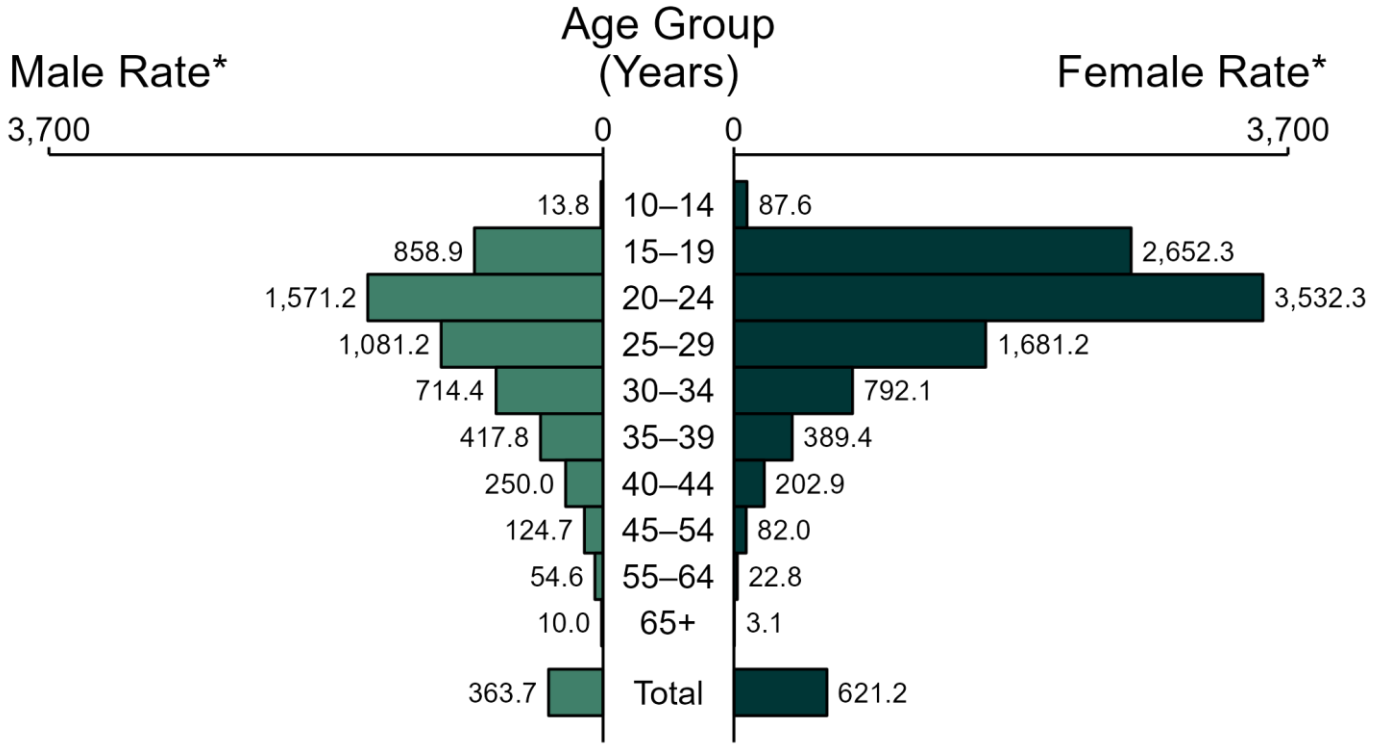
NOTE: "Total syphilis" includes all stages of syphilis and congenital syphilis.

Chlamydia — Rates of Reported Cases by Jurisdiction, United States and Territories, 2022



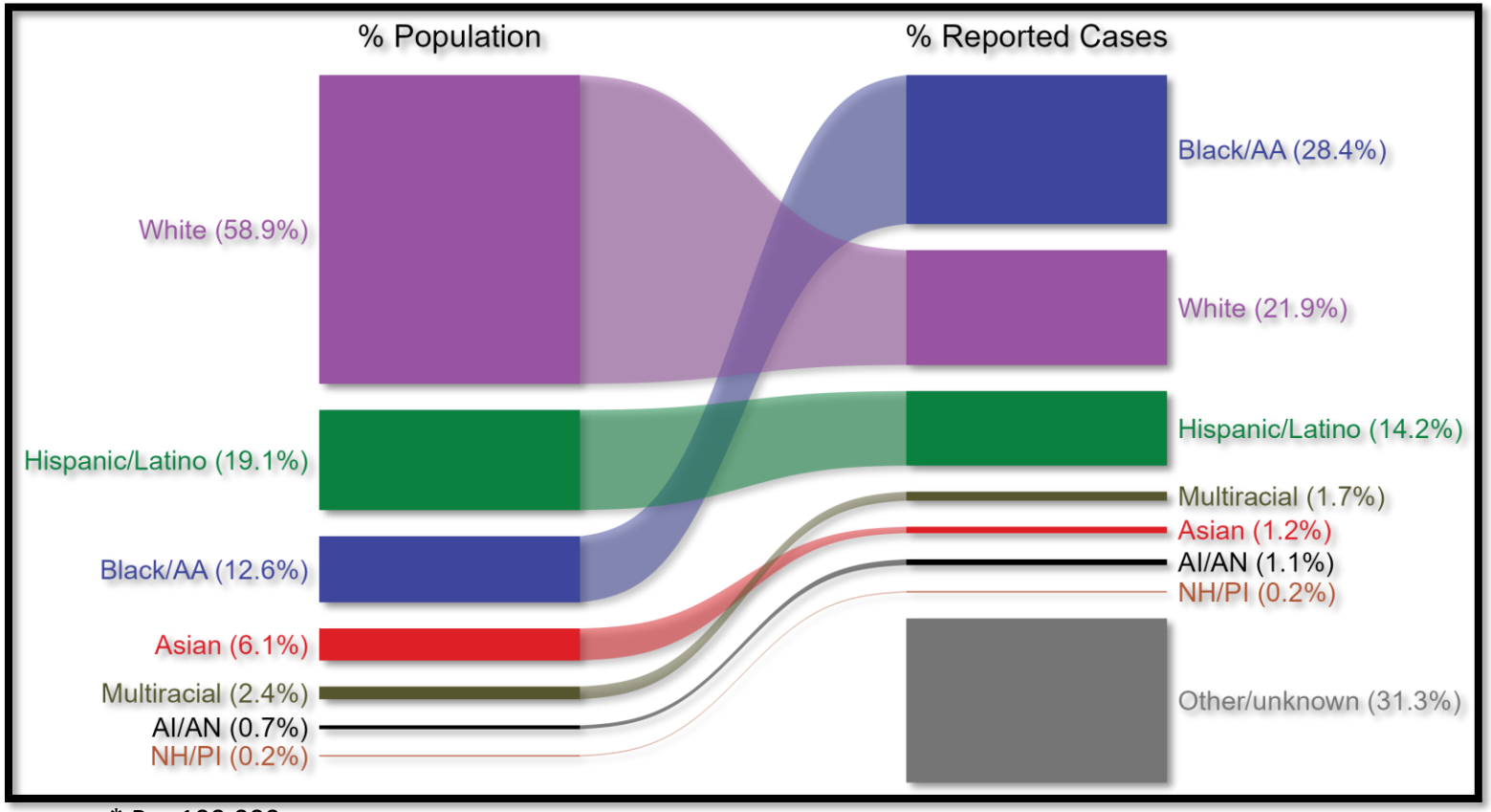
- * Per 100,000

Chlamydia — Rates of Reported Cases by Age Group and Sex, United States, 2022



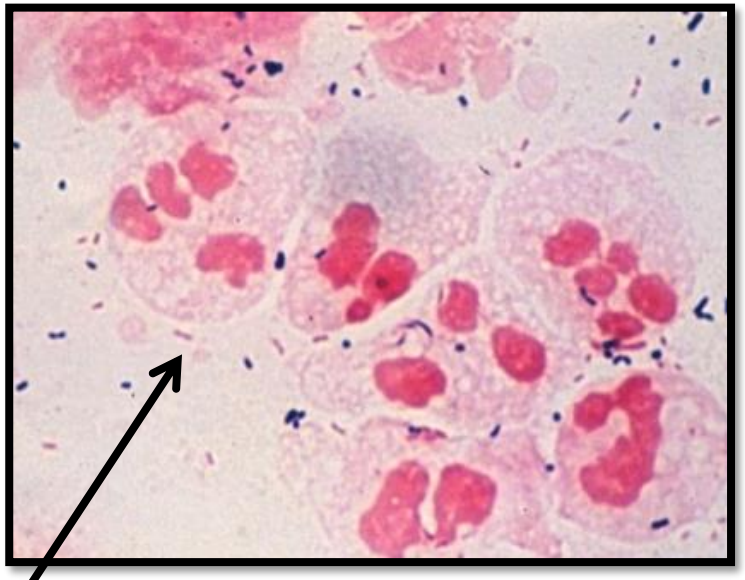
- * Per 100,000
- **NOTE:** Total includes cases of all ages, including those with unknown age.

Chlamydia — Total Population and Reported Cases by Race/Hispanic Ethnicity, United States, 2022

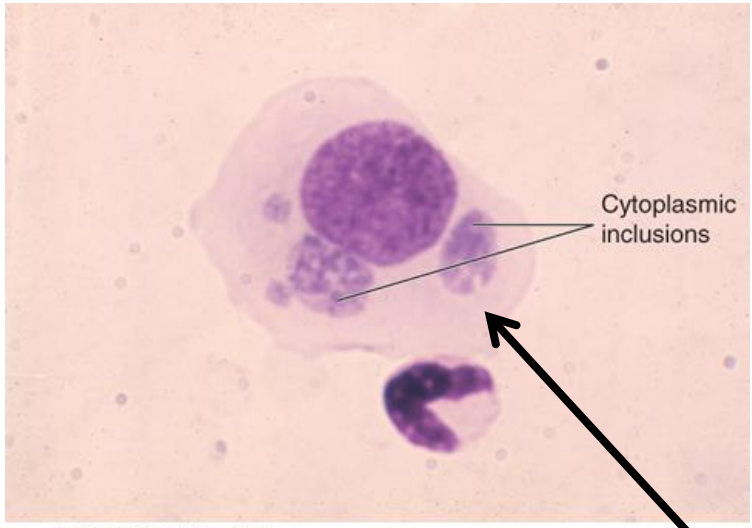


- * Per 100,000
- **NOTE:** In 2022, a total of 515,552 chlamydia cases (31.3%) had missing, unknown, or other race and were not reported to be of Hispanic ethnicity. These cases are included in the "other/unknown" category.
- **ACRONYMS:** AI/AN = American Indian or Alaska Native; Black/AA = Black or African American; NH/PI = Native Hawaiian or other Pacific Islander

Chlamydia: Bacteriology



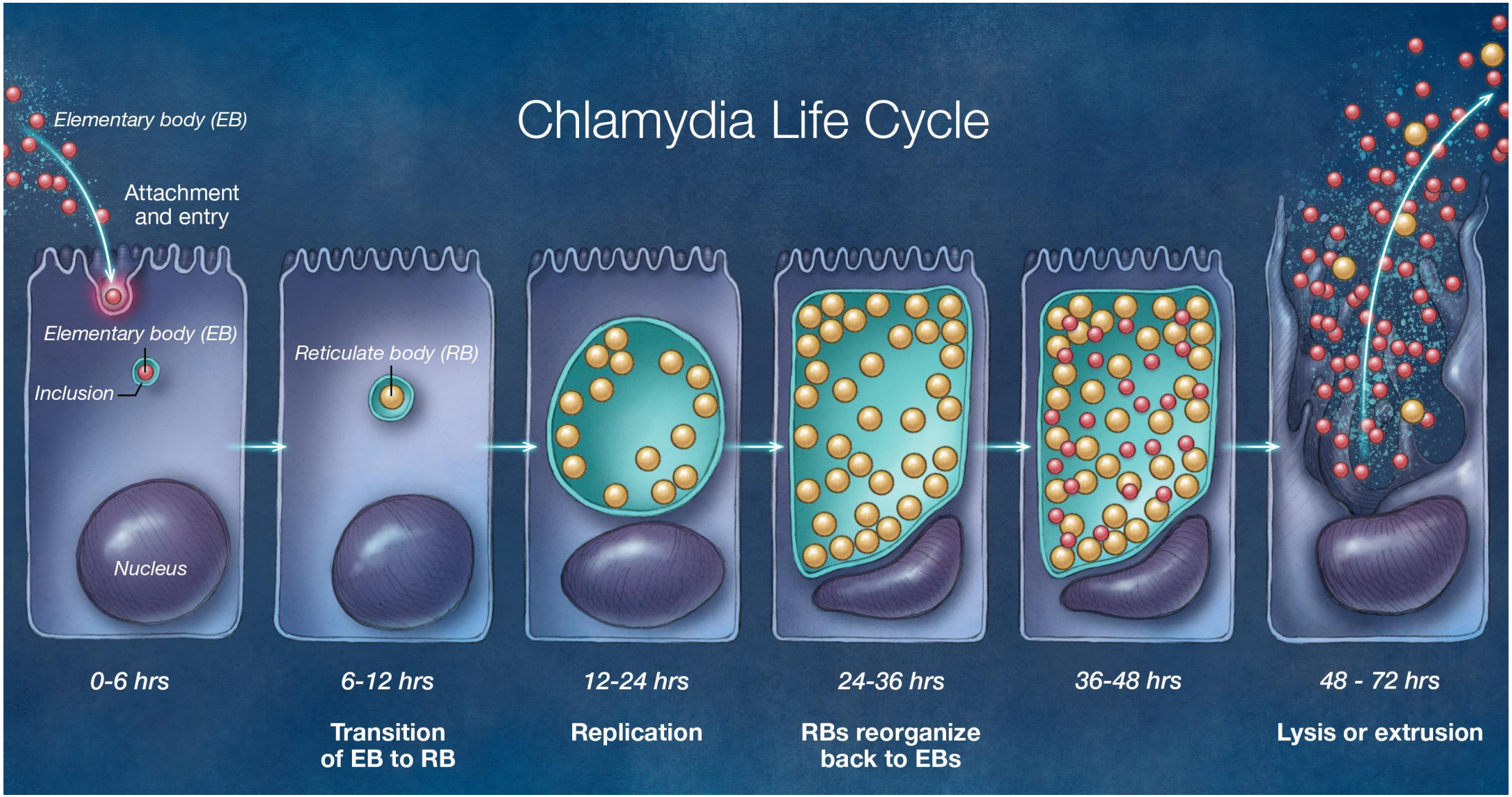
- Weakly Gram-negative, intracellular bacteria that are typically rod-shaped



Source: Kenneth J. Ryan
Sherris & Ryan's Medical Microbiology, 8e
Copyright © McGraw Hill. All rights reserved.

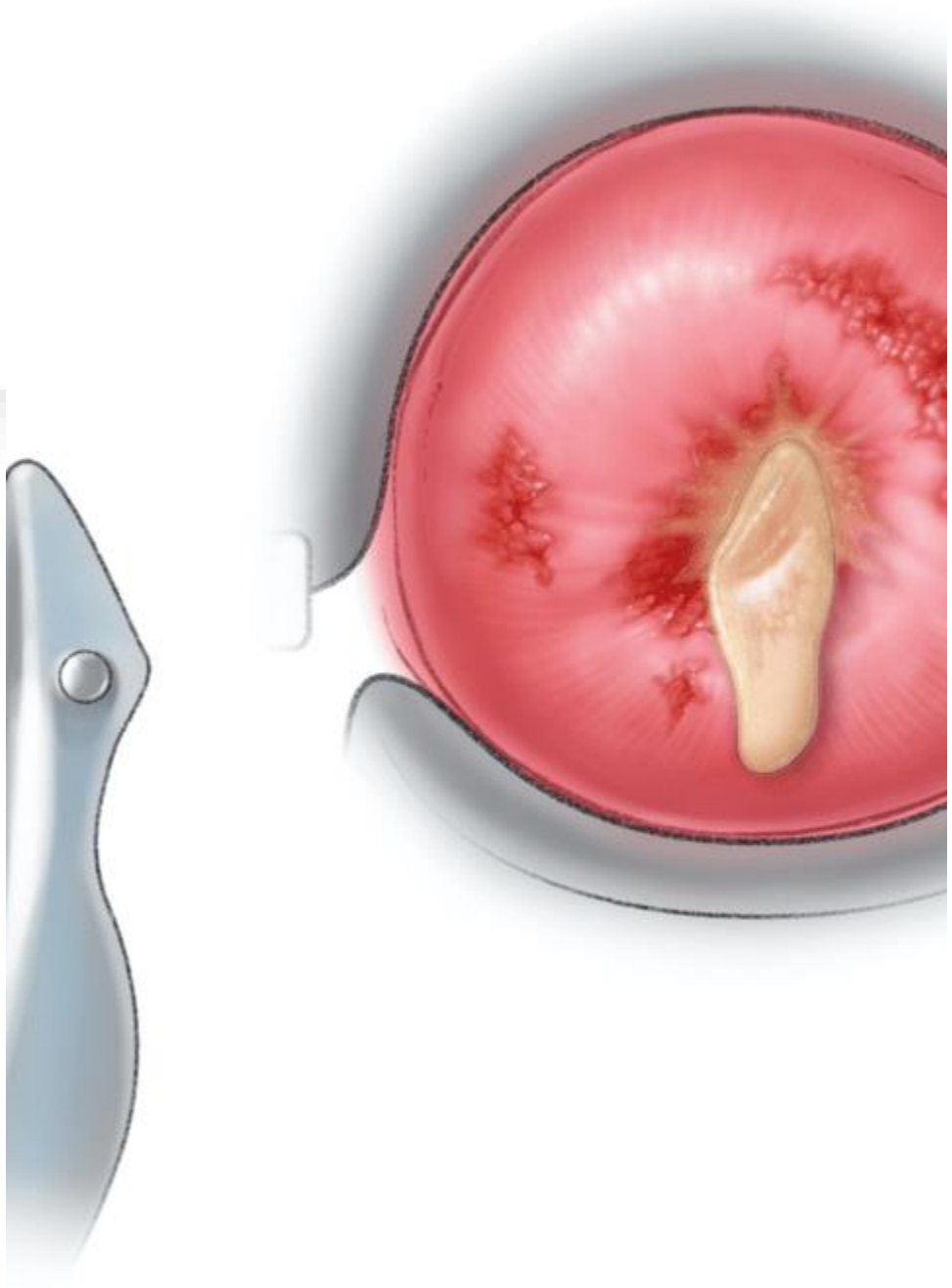
- lacks a cell wall

Chlamydia: Life Cycle



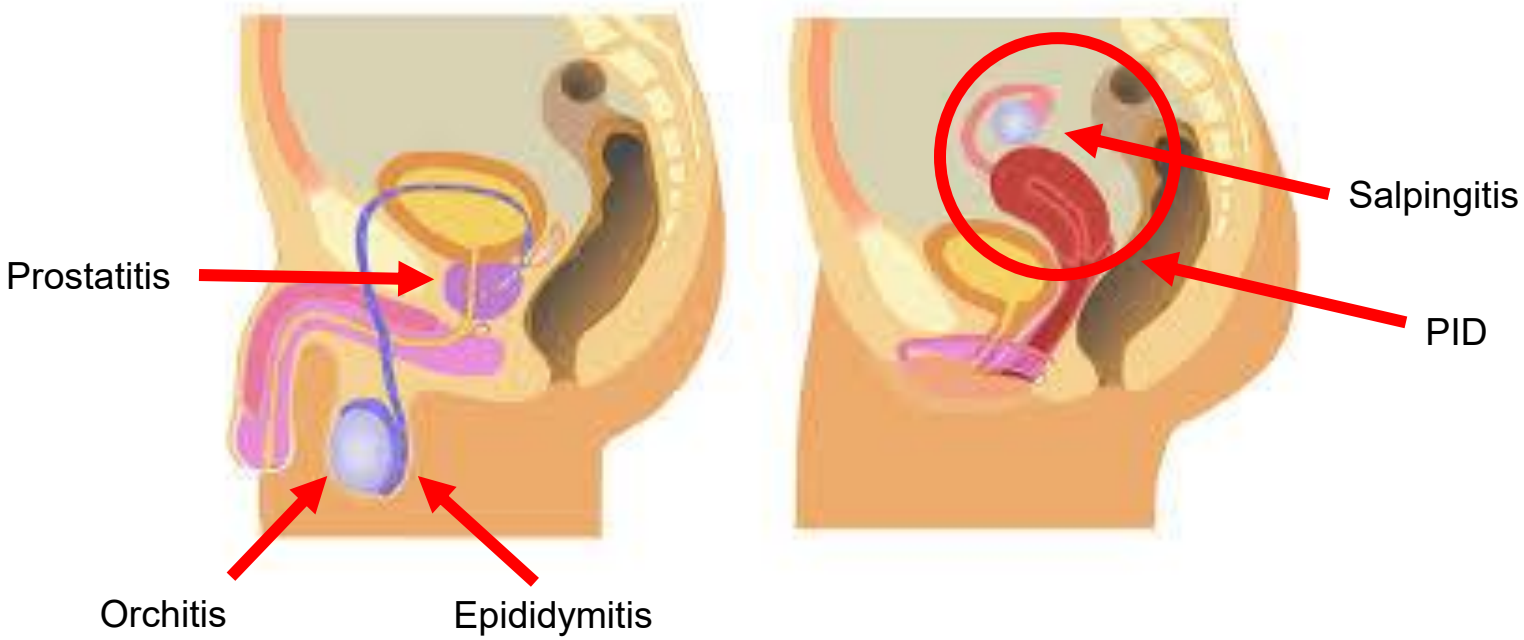
Chlamydia: Presentation

- Clinical syndromes may include:
  - **Nongonococcal urethritis**: may have pain with urination and thin to thick mucoid discharge from the urethra
  - **Cervicitis**: with inflammation of the cervix, often with red appearance on exam with yellow mucopurulent discharge



Chlamydia: Presentation

- Infection can also **ascend the genital tract** and lead to:



Chlamydia: Presentation

- Other sites **outside of the genital tract** can be involved including:
  - Pharyngitis (oral sex)
  - Proctitis (anal sex)
  - Conjunctivitis (Trachoma) *leading cause of blindness in world
  - Perihepatitis (Fitz-Hugh-Curtis Syndrome)
  - Reactive arthritis (triad of conjunctivitis, urethritis, and arthritis)

Chlamydia: Trachoma





Chlamydia: Presentation

- Patients can also be **ASYMPTOMATIC**
  - especially biologic females
  - leads to continued transmission
  - may miss or delay diagnosis
  - increases risk of developing PID

Chlamydia: Diagnosis

- **Who To Test**:
  - Patients who are symptomatic
  - Recent exposure or other STI diagnosis
  - Asymptomatic screening for:
    - women who have sex with men
    - men who have sex with men
    - Pregnancy
    - persons living with HIV
    - Persons on PrEP and/or doxy PEP

Chlamydia: Diagnosis

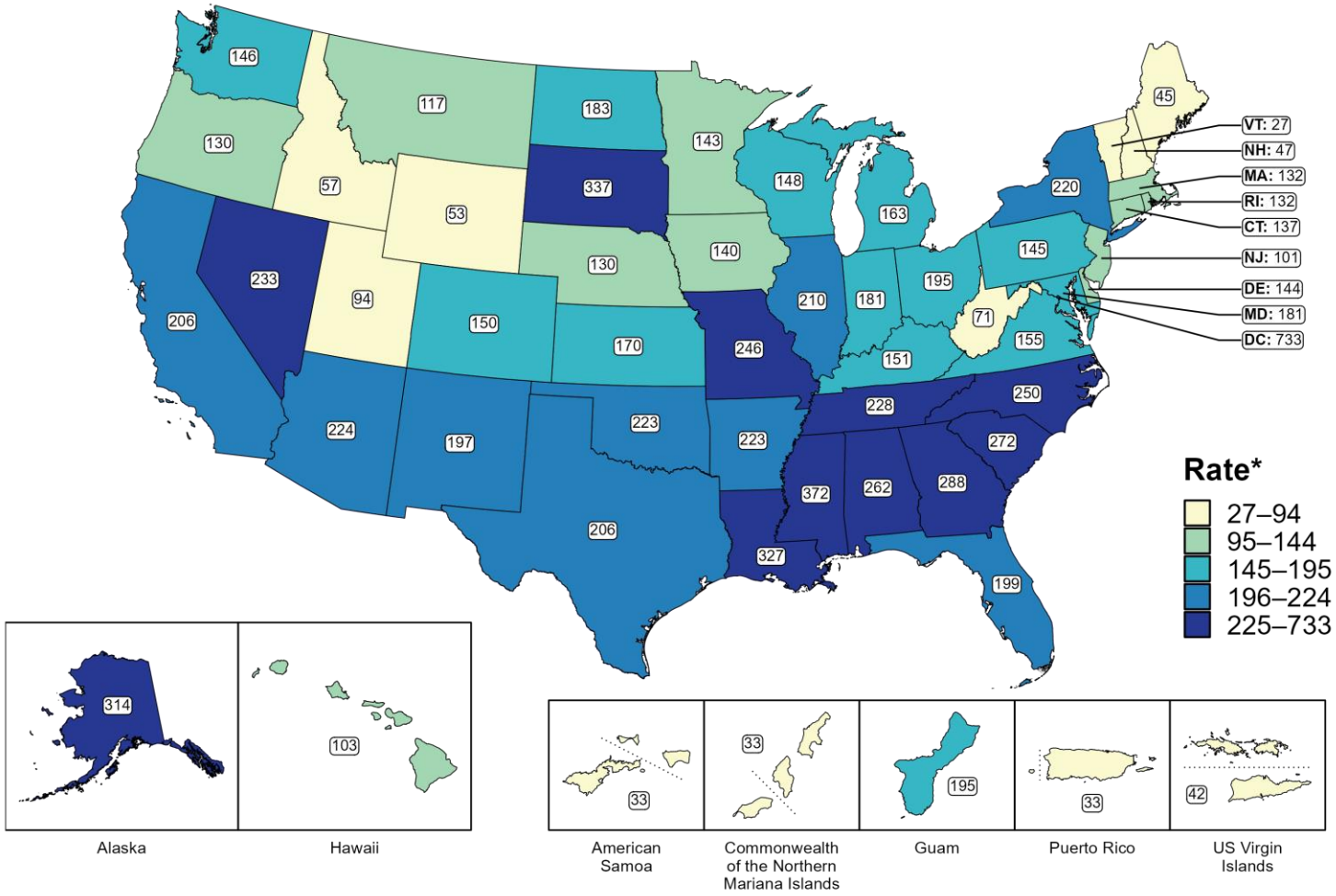
- How To Test:
- Nucleic Acid Amplification test (**NAAT**)
  - –Preferred
  - – can be run on vaginal or cervical specimens, urethral specimens, urine, oropharyngeal, and rectal specimens
  - –Sensitivity 97-100%
- Culture
- Serology

Chlamydia: Treatment

- **doxycycline 100 mg po bid x 7 days**
  - – Alternative azithromycin 1g po x 1 dose or levofloxacin 500 mg po daily x 7 days
  - –In pregnancy, amoxicillin 500 mg po tid x 7 days
  - –No resistance has been described
  - – Consider test of cure if continued symptoms, pregnant, or suboptimal treatment

Neisseria gonorrhoeae

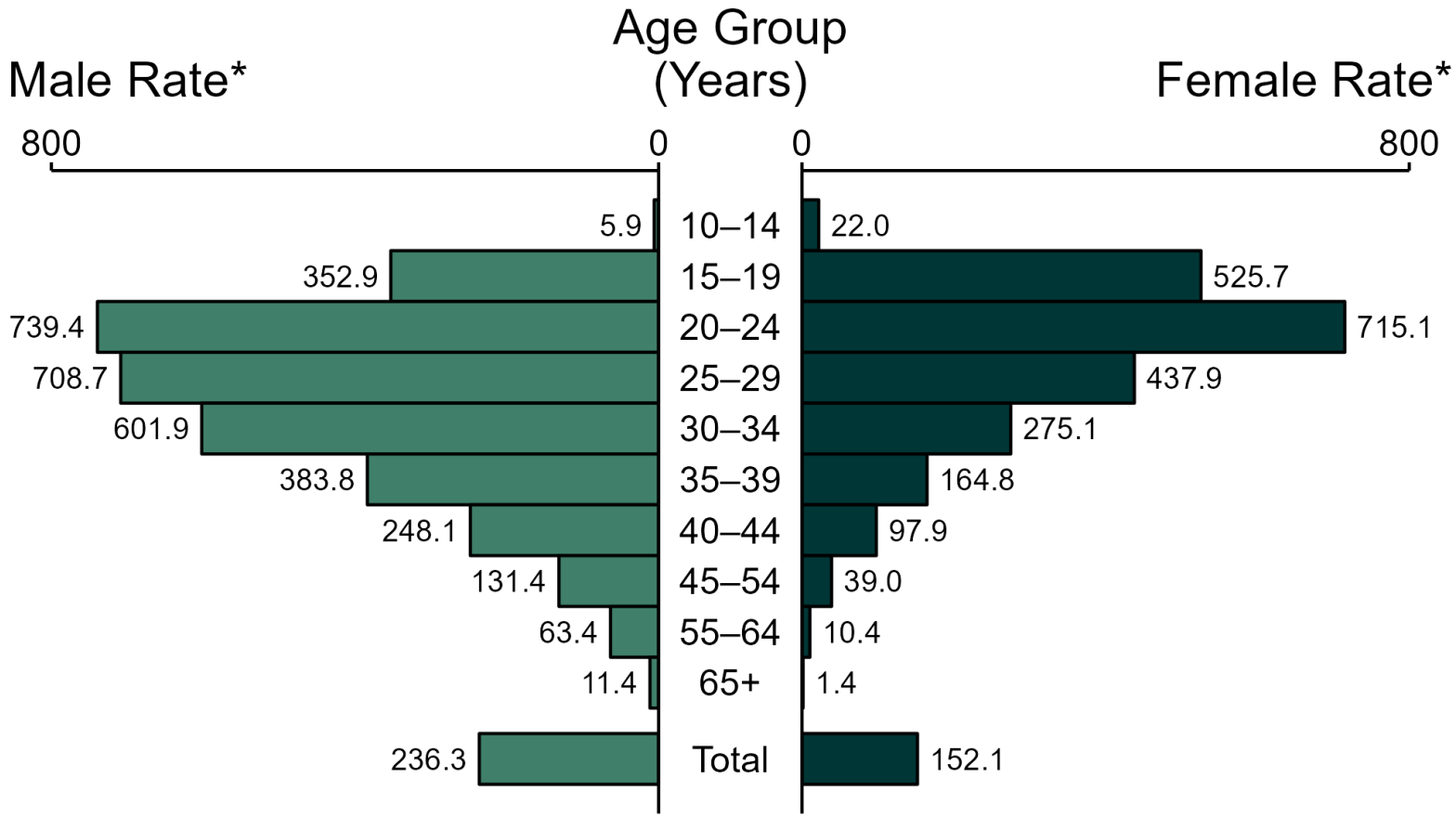
Gonorrhea — Rates of Reported Cases by Jurisdiction, United States and Territories, 2022



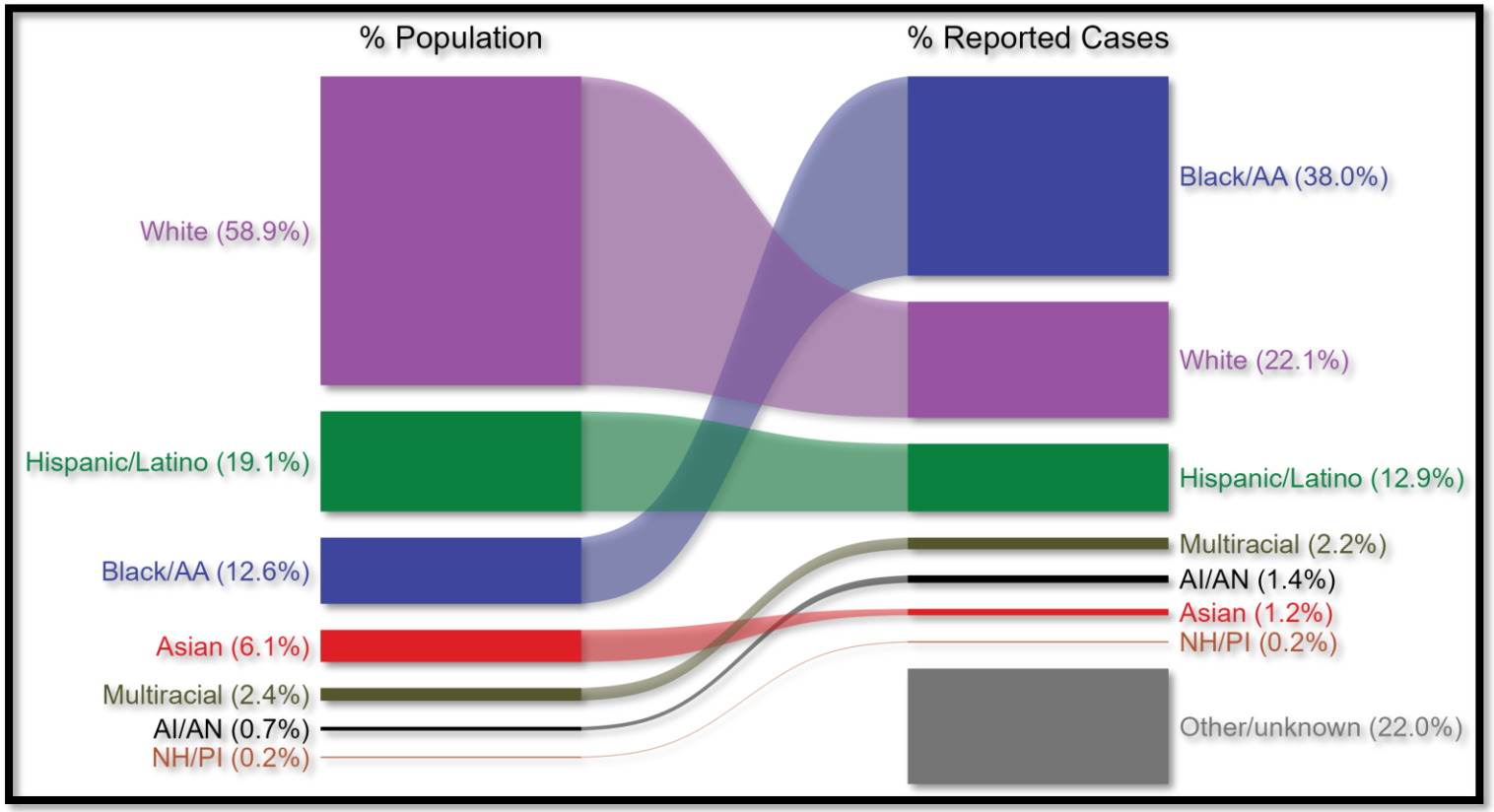
- * Per 100,000

Gonorrhea — Rates of Reported Cases by Age Group and Sex, United States, 2022

- * Per 100,000
- **NOTE:** Total includes cases of all ages, including those with unknown age.

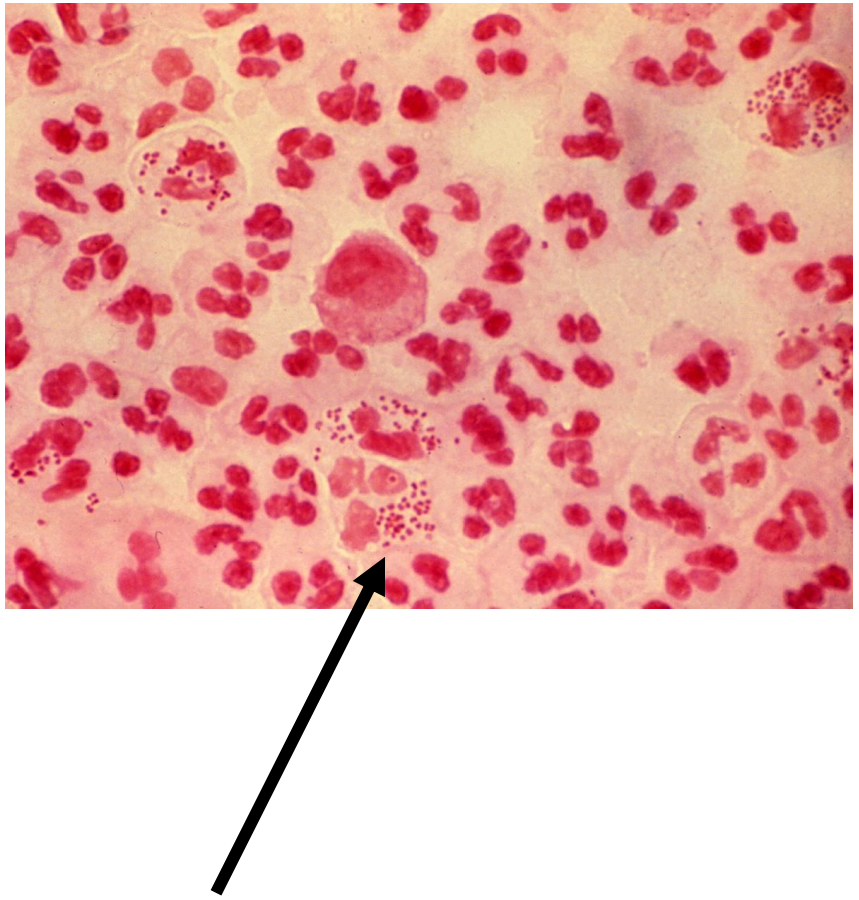


Gonorrhea — Total Population and Reported Cases by Race/Hispanic Ethnicity, United States, 2022



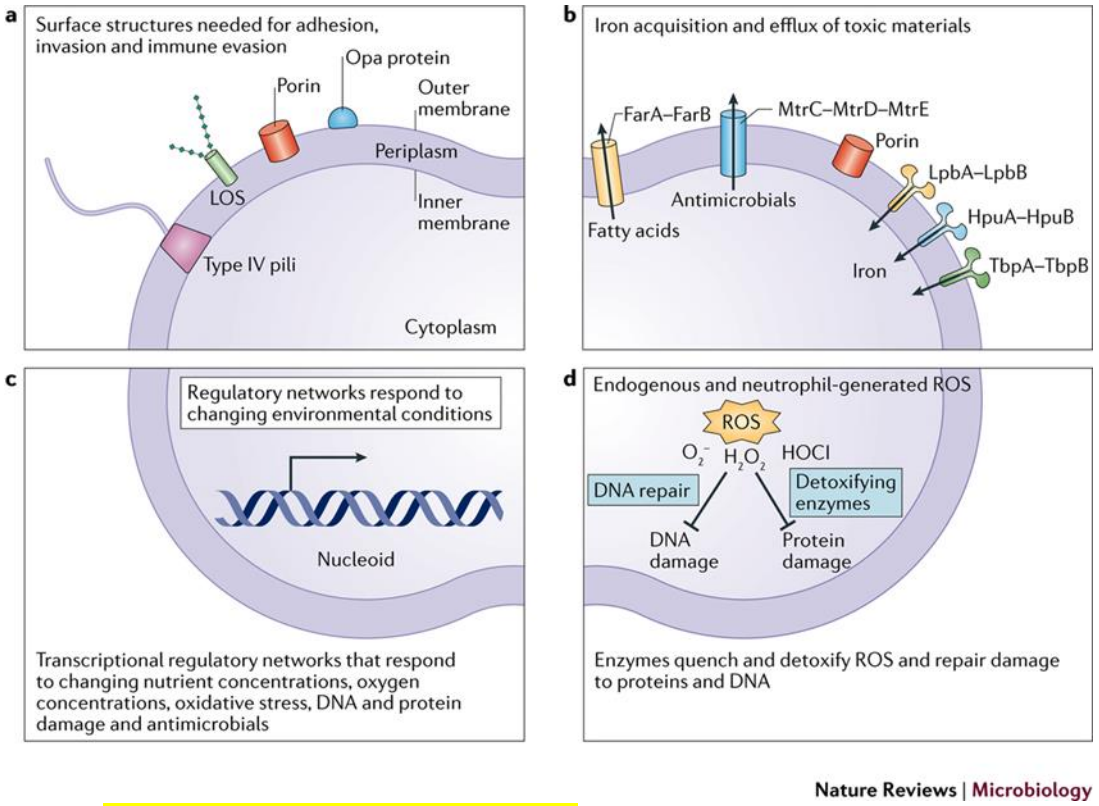
- * Per 100,000
- **NOTE:** In 2022, a total of 142,317 gonorrhea cases (22.0%) had missing, unknown, or other race and were not reported to be of Hispanic ethnicity. These cases are included in the "other/unknown" category.
- **ACRONYMS:** AI/AN = American Indian or Alaska Native; Black/AA = Black or African American; NH/PI = Native Hawaiian or other Pacific Islander

Gonorrhea: Bacteriology



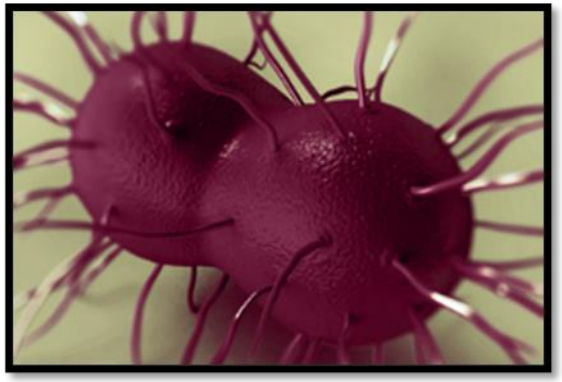
- Gram-negative diplococci
  (divides by binary fission)

Gonorrhea: Bacteriology



Nature Reviews | Microbiology

- Lipo-oligo saccharide (LOS) – which is different than the lipo-polysaccharide (LPS) of gram-negative bacilli. The LOS incorporates host sialic acid onto its surface to help evade the immune system



- Pili are virulence factors that help the bacteria attach to and invade the host mucosa
- The genes that encode the pili undergo frequent recombination, making it difficult for the body to develop immunity to *N. gonorrhoeae* and also for vaccine targets

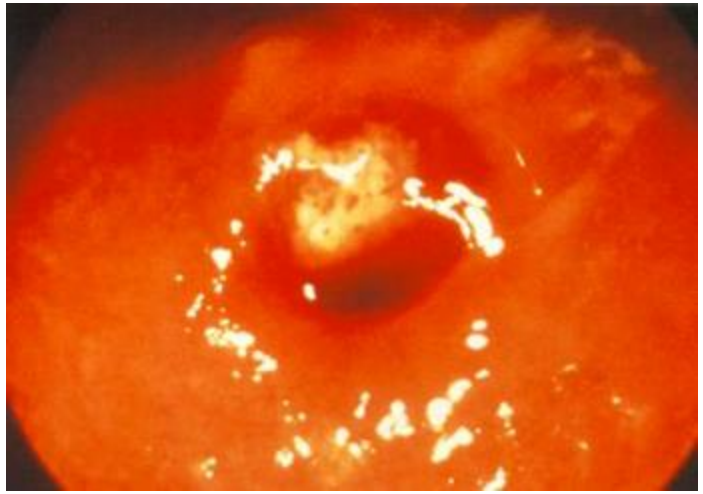
Gonorrhea: Presentation

- Clinical syndromes in adults include:
  - **Urethritis**: may have pain with urination and thick purulent discharge from the urethra
  - **Cervicitis**: with inflammation of the cervix, often with red appearance on exam with yellow mucopurulent discharge

Gonorrhea: Presentation



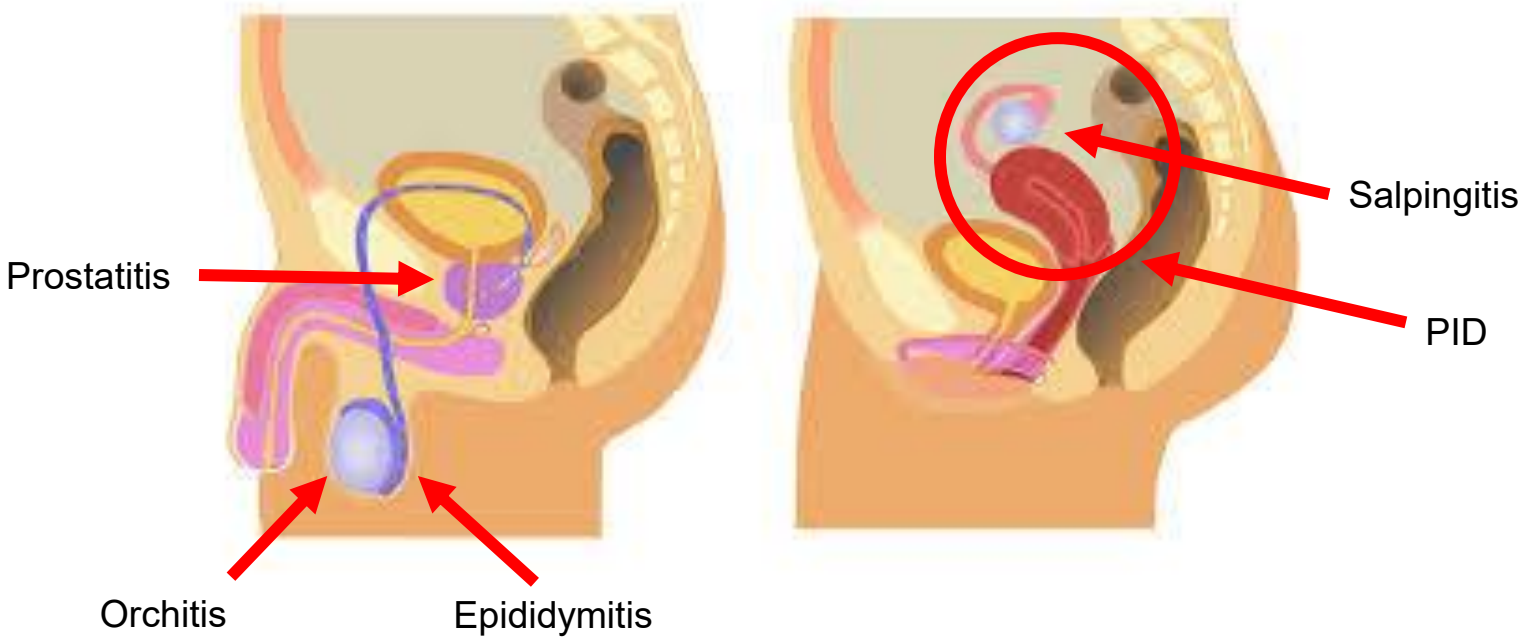
Urethritis



Cervicitis

Gonorrhea: Presentation

- Infection can also **ascend the genital tract** and lead to:



Gonorrhea: Presentation

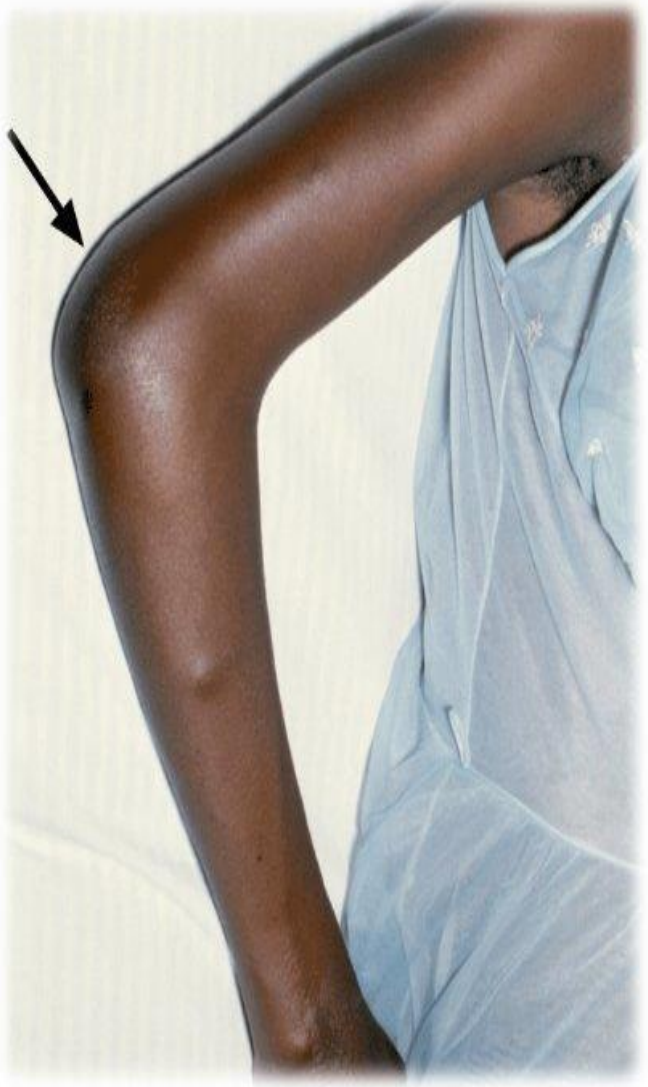
- Other sites **outside of the genital tract** can be involved including:
  - Pharyngitis
  - Proctitis
  - Conjunctivitis
  - Bacteremia
  - Septic arthritis
  - Disseminated (<3%), typically characterized by fever, malaise, polyarthralgias, tenosynovitis, and dermatitis.

Gonorrhea: Pharyngitis



Source: Nanda Rachmad Putra Gofur, Aisyah Rachmadani Putri Gofur, Soesilaningtyas, Rizki Nur Rachman Putra Gofur, Mega Kahdina, et al. Resistance of Neisseria gonorrhoeae on oral and mucosa : A Review Article. Prog Asp in Pediatric & Neonat 3(2)- 2021. PAPN. MS.ID.000158. DOI: 10.32474/PAPN.2021.03.000158

Gonorrhea: Disseminated





Gonorrhea: Presentation

- Patients can also be **ASYMPTOMATIC**
  - especially biologic females
  - leads to continued transmission
  - may miss or delay diagnosis
  - increases risk of developing PID

Gonorrhea: Diagnosis

- Who To Test: Same as for Chlamydia
  - Symptomatic, exposure, pregnancy
  - Asymptomatic
  - DO NOT forget about extra-genital screening
    - MSM: up to 65% of GC cases may be missed if genital only screening is done
    - Biologic females: up to 31% may be missed

Gonorrhea: Diagnosis

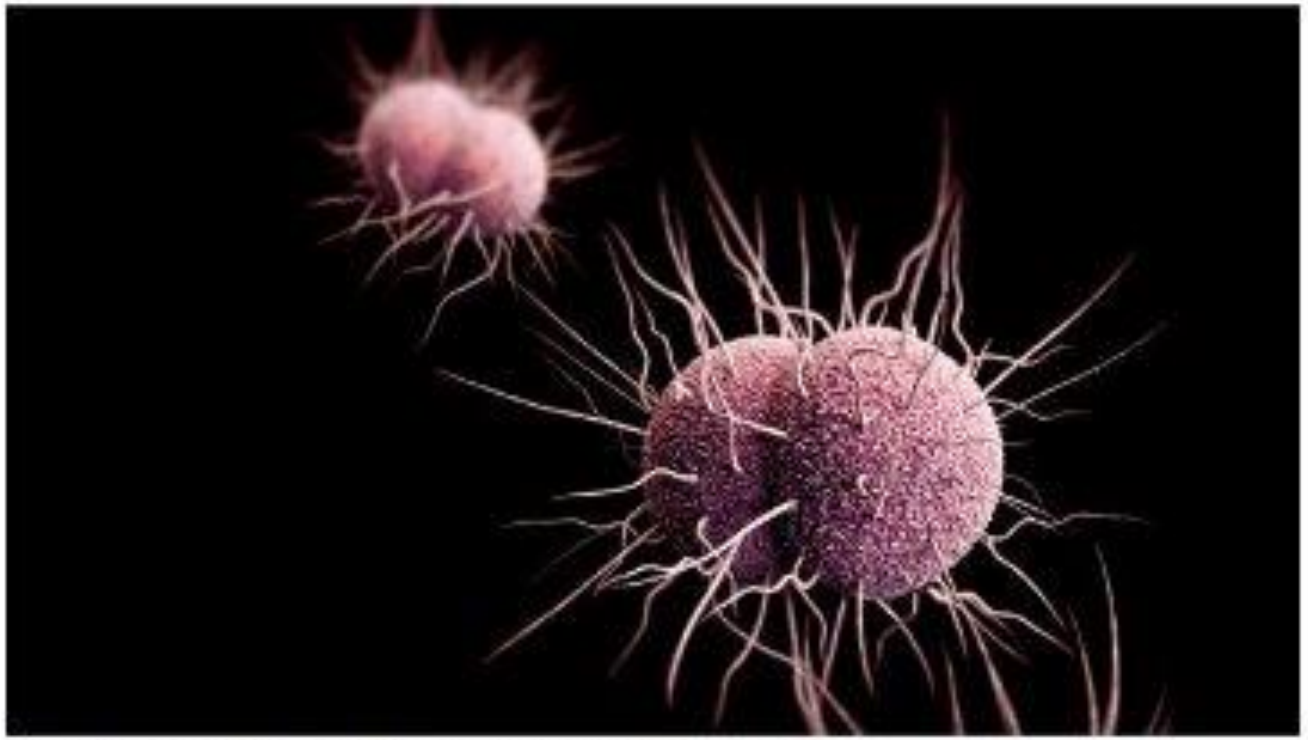
- How To Test: Same as for Chlamydia
  - **NAAT**
    - Vaginal swab preferred, can also use urine and cervical specimens
    - Urine specimen preferred, urethral swab alternative
    - Rectal and pharyngeal swab may be used for extra-genital screening (though not approved by FDA)
    - Sensitivity 95-100%
  - Culture
    - Urethral swab and endocervical specimens
    - Sensitivity much lower for pharyngeal and rectal samples (<50%)

Gonorrhea: Treatment

- **Ceftriaxone 500 mg IM x 1 dose**
  - **Also treat empirically for chlamydia** with doxycycline 100 mg po bid x 7 days
  - Alternate if true allergy to cephalosporins: 240 mg IM gentamicin plus azithromycin 2g po x1 dose

By ASHLEY WELCH | CBS NEWS | March 29, 2018, 5:10 PM

"Huge concern" about spread of super-resistant gonorrhea



Medical illustration of gonorrhea bacteria Neisseria gonorrhoeae | CDC

Pelvic Inflammatory Disease (PID)

PID

- Definition:
  - **infection of higher up structures of the reproductive tract** including uterus (endometritis), fallopian tubes (salpingitis) and/or ovaries (oophoritis)

PID

- Main pathogens:
  - *Chlamydia trachomatis*
  - *Neisseria gonorrhoeae*

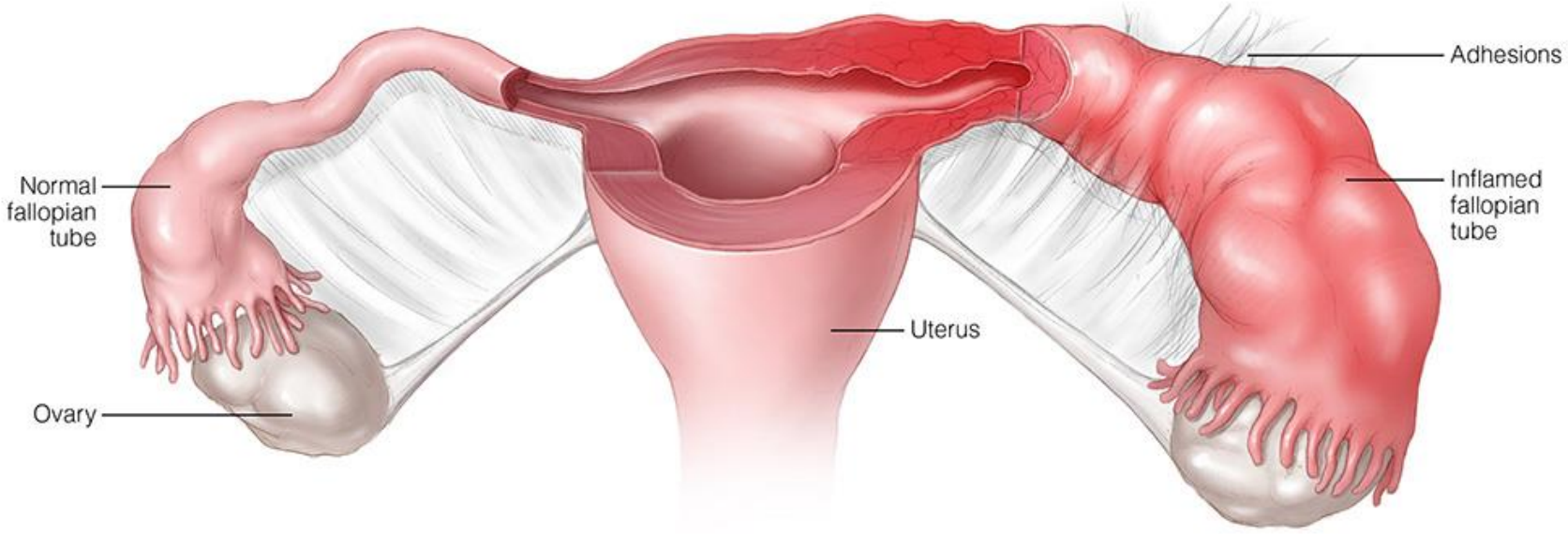
PID: Presentation

- Clinical manifestations may include:
  - **Fever**
  - **Lower abdominal pain**
  - **Abnormal menstrual bleeding**
  - **Cervical motion tenderness**
    - “Chandelier sign”

PID: Complications

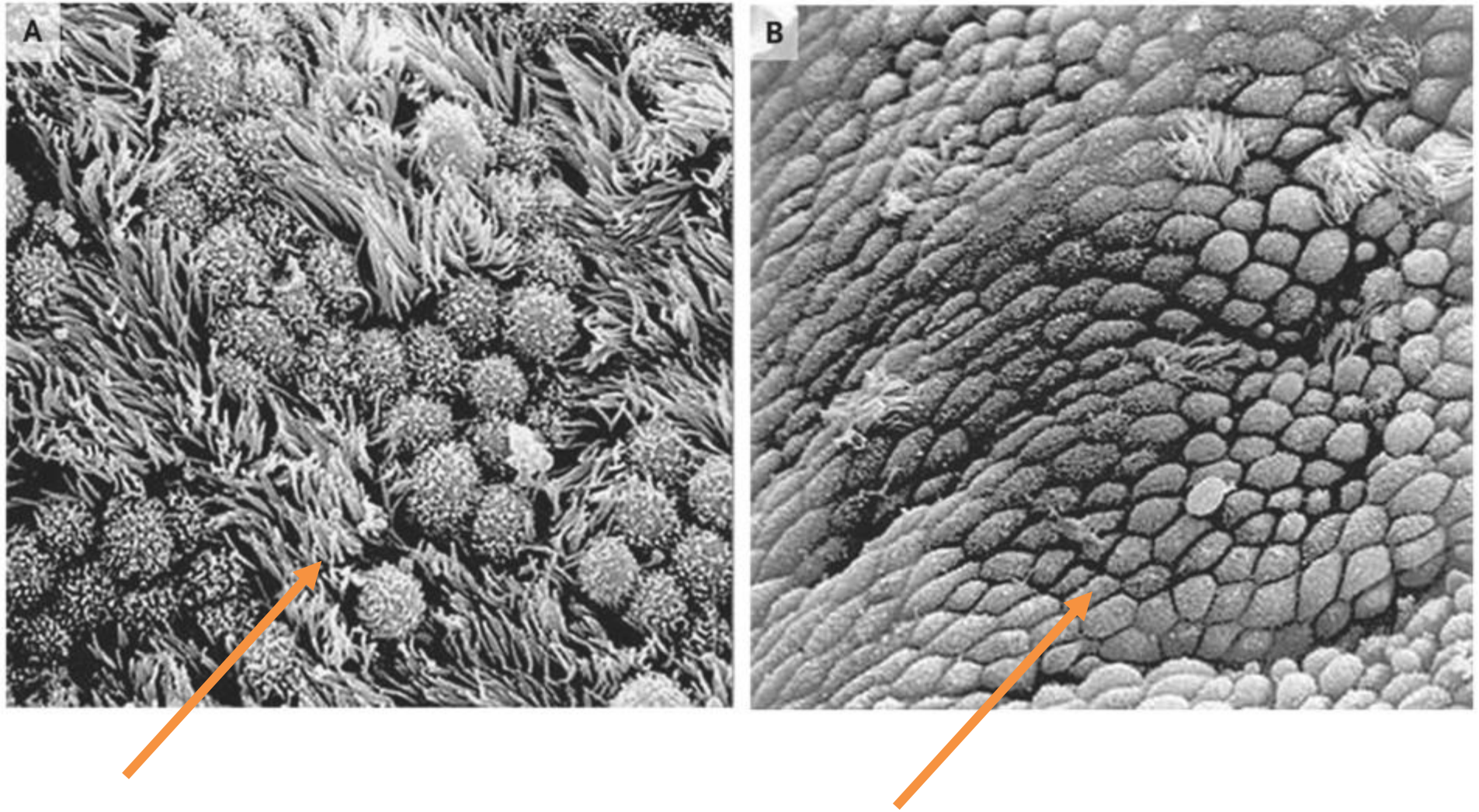
- **Complications** are significant and include:
  - Scarring of the fallopian tubes, which can lead to sterility or ectopic pregnancy
  - Abscesses
  - Peritonitis
  - Perihepatitis (Fitz-Hugh-Curtis syndrome)

PID: Complications



- Sterility caused by scarring of fallopian tubes preventing sperm from reaching the egg
- Ectopic pregnancy also the result of scarring of the fallopian tubes, inhibiting egg transit to uterus for implantation
- Peri-hepatitis causes infection of the capsule of the liver, patients typically present with RUQ pain ans tenderness

PID: Complications



Brunham RC, Gottlieb SL, Paavonen J. Pelvic inflammatory disease. N Engl J Med. 2015;372:2039-48.

PID: Complications

- **Perihepatitis** (Fitz-Hugh-Curtis Syndrome)
  - Right upper quadrant abdominal pain due to inflammation of the liver capsule and local peritoneum
  - adhesions are formed
  - may have referred pain to right shoulder
  - difficult to diagnose as genital symptoms usually minimal ands labs may be normal, must have high clinical suspicion

PID: Complications

- **Perihepatitis** (Fitz-Hugh-Curtis Syndrome)



PID: Diagnosis

- Acute PID is often **difficult to diagnose**

- Diagnosis should be considered in sexually-active patients who present with lower abdominal pain without alternative etiology and one of the following:
  - Cervical motion tenderness
  - Uterine tenderness
  - Adnexal tenderness

PID: Treatment

- Parenteral/ intramuscular ceftriaxone plus po doxycycline with or without metronidazole x 14 days

- Alternative oral:
  - Levofloxacin + metronidazole
  - Moxifloxacin

Syphilis:
Treponema pallidum

Syphilis: Bacteriology

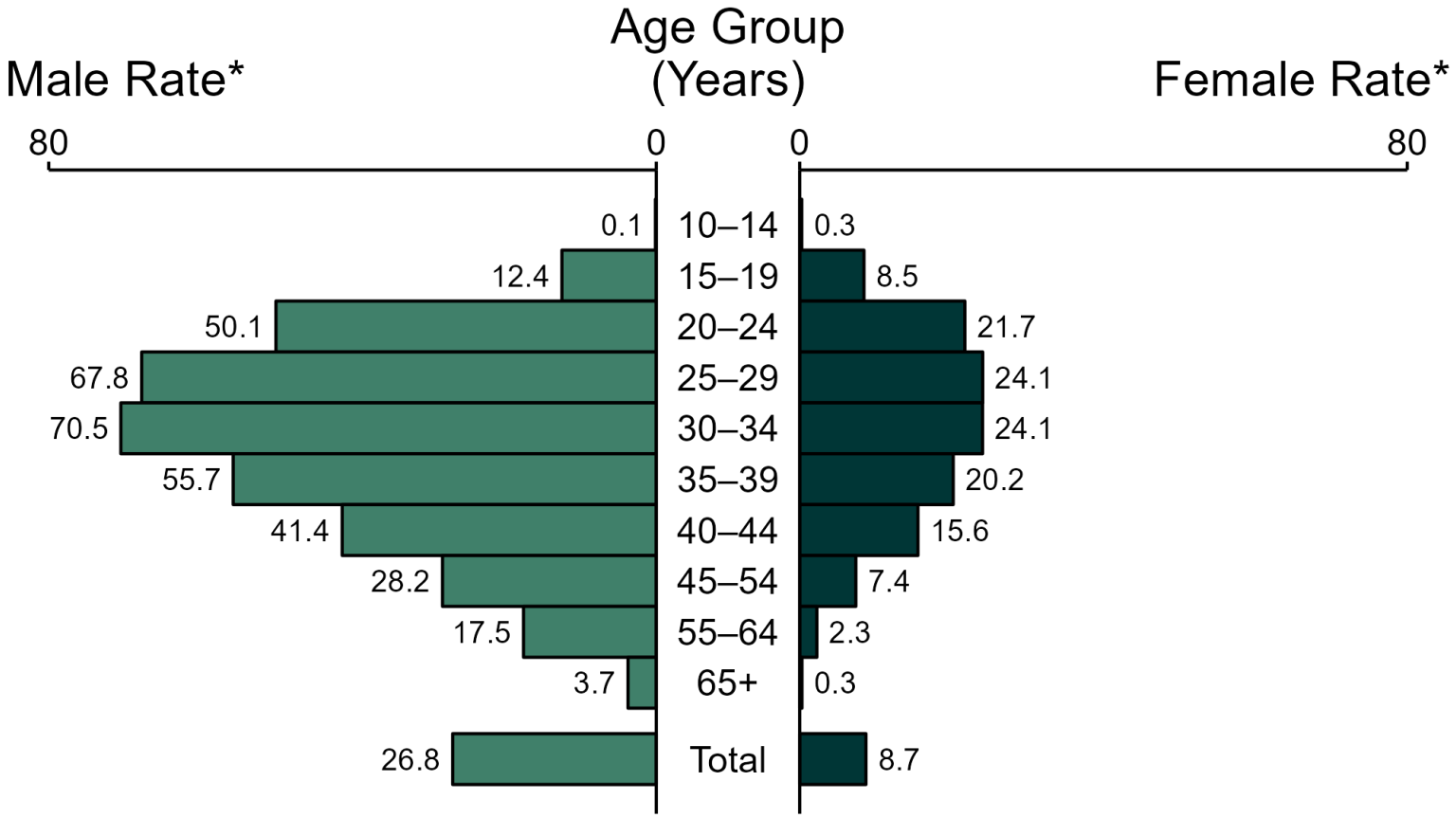




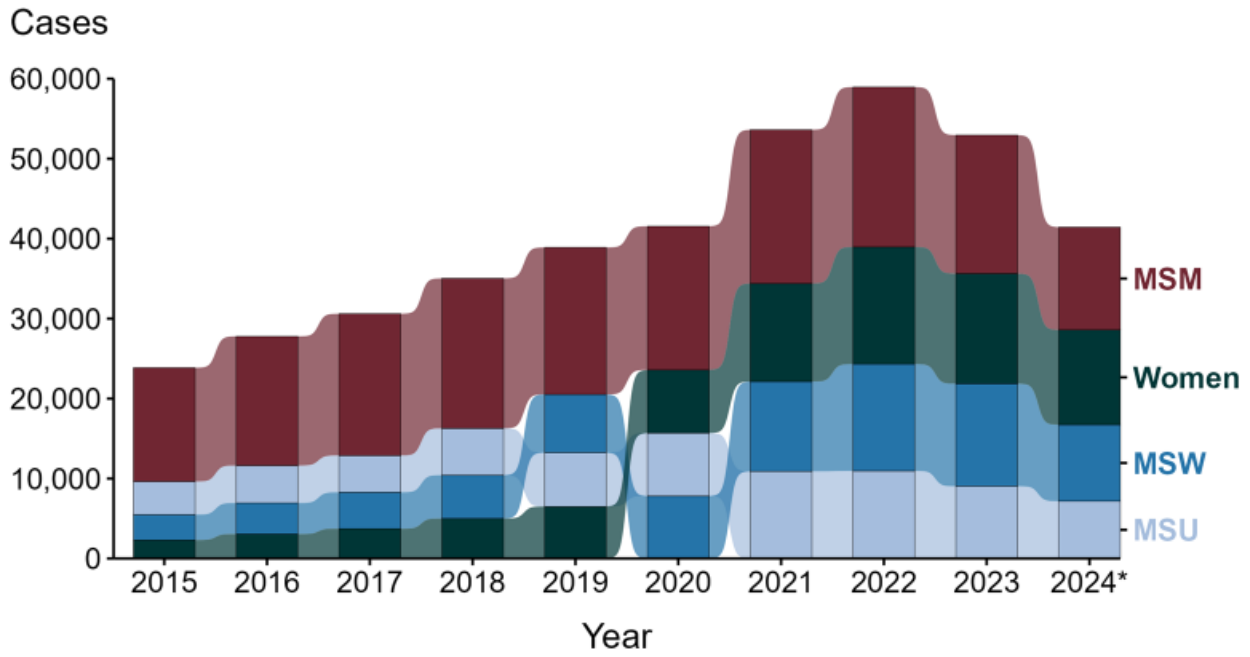
- Gram negative, corkscrew-shaped bacteria
  - –Too small to see on gram stain!
  - –Too picky to grow on routine culture!

Primary and Secondary Syphilis — Rates of Reported Cases by Age Group and Sex, United States, 2022

- * Per 100,000
- **NOTE:** Total includes cases of all ages, including those with unknown age.



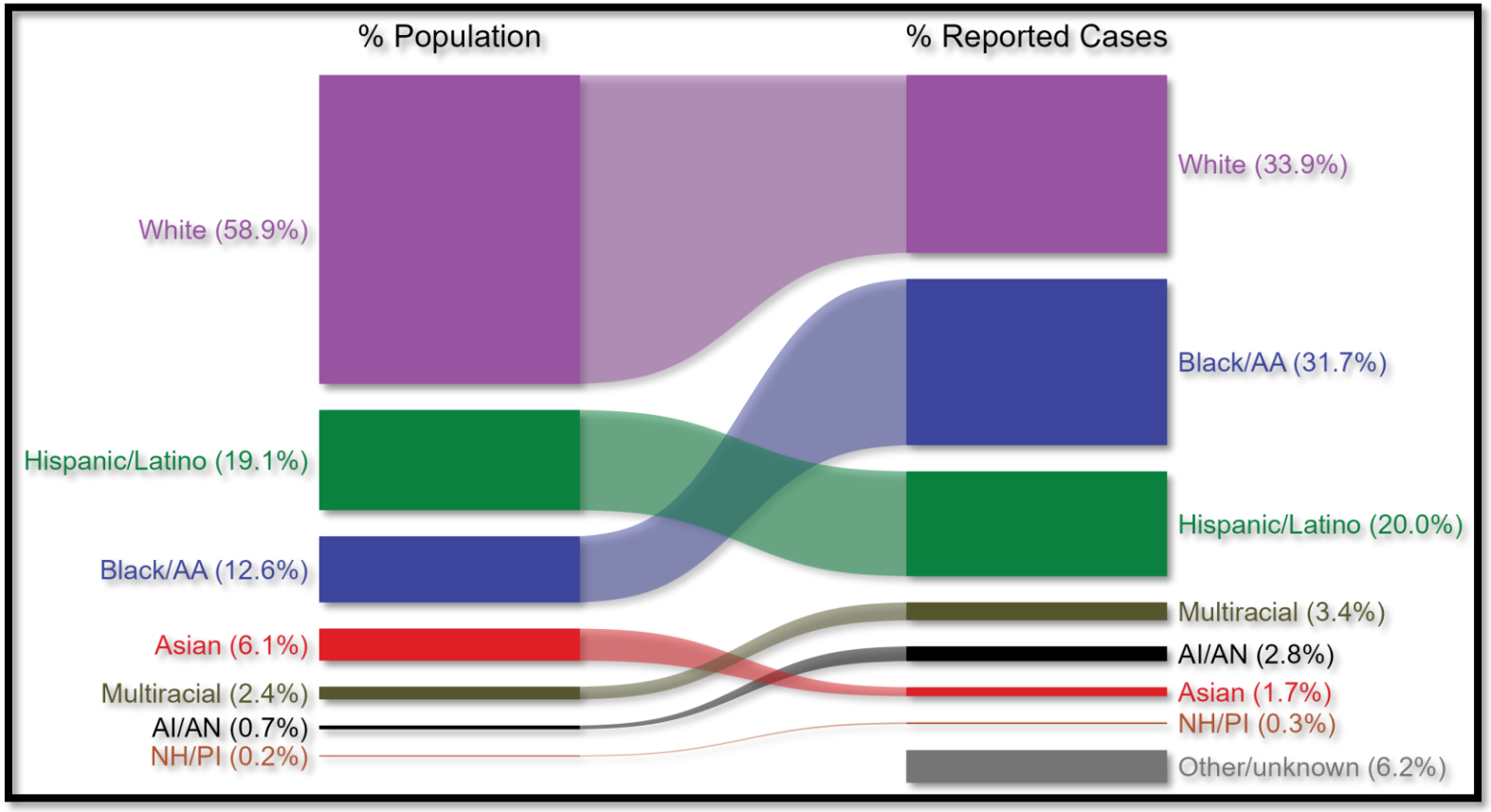
Primary and Secondary Syphilis — Reported Cases by Sex and Sex of Sex Partners and Year, United States, 2015–2024



* 2024 data are provisional as of August 14, 2025.

ACRONYMS: MSM = Men who have sex with men; MSU = Men with unknown sex of sex partners; MSW = Men who have sex with women only

Primary and Secondary Syphilis — Total Population and Reported Cases by Race/Hispanic Ethnicity, United States, 2022



- * Per 100,000
- **NOTE:** In 2022, a total of 3,686 primary and secondary (P&S) syphilis cases (6.2%) had missing, unknown, or other race and were not reported to be of Hispanic ethnicity. These cases are included in the "other/unknown" category.
- **ACRONYMS:** AI/AN = American Indian or Alaska Native; Black/AA = Black or African American; NH/PI = Native Hawaiian or other Pacific Islander

Syphilis: Presentation

- Clinical manifestations/ stages:
  - **Primary**: one or more painless ulcerative lesions (chancre)
  - **Secondary**: localized or diffuse rash with LAD and other associated symptoms
  - **Tertiary**: late-stage organ manifestations
  - **Latent** (asymptomatic)

Primary syphilis-chancre







Secondary syphilis-rash







Secondary syphilis-mucous patch



A



B

Secondary syphilis-condyloma lata



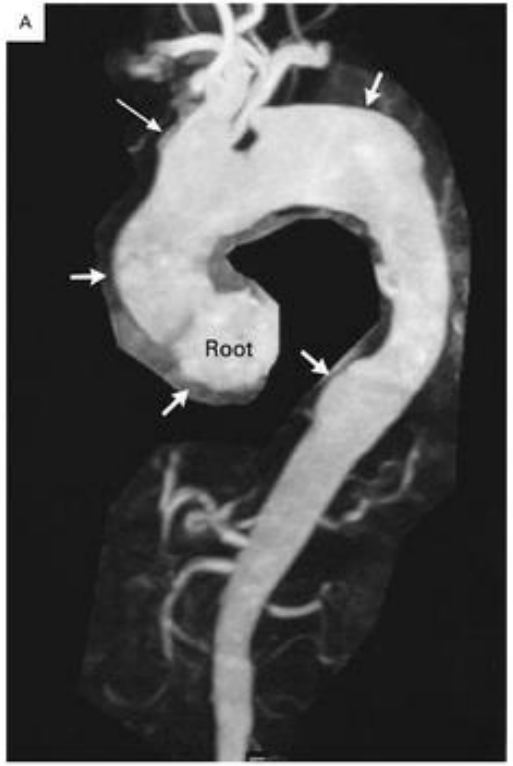
Tertiary syphilis

gumma





Argyll-Roberston pupil





Aortic aneurysm

Syphilis: Presentation

- Latent infection: ASYMPTOMATIC but positive test
  - **Early**: infection is suspected to be <12 months
  - **Late**: infection is suspected to be >12 months or is unknown

**Important to distinguish because infectivity and treatment vary between these 2 stages*

Syphilis: Diagnosis

- Who To Test:
  - Patients who are symptomatic
  - Asymptomatic screening:
    - Pregnancy
    - MSM
    - Persons living with HIV
    - PrEP or doxy PEP

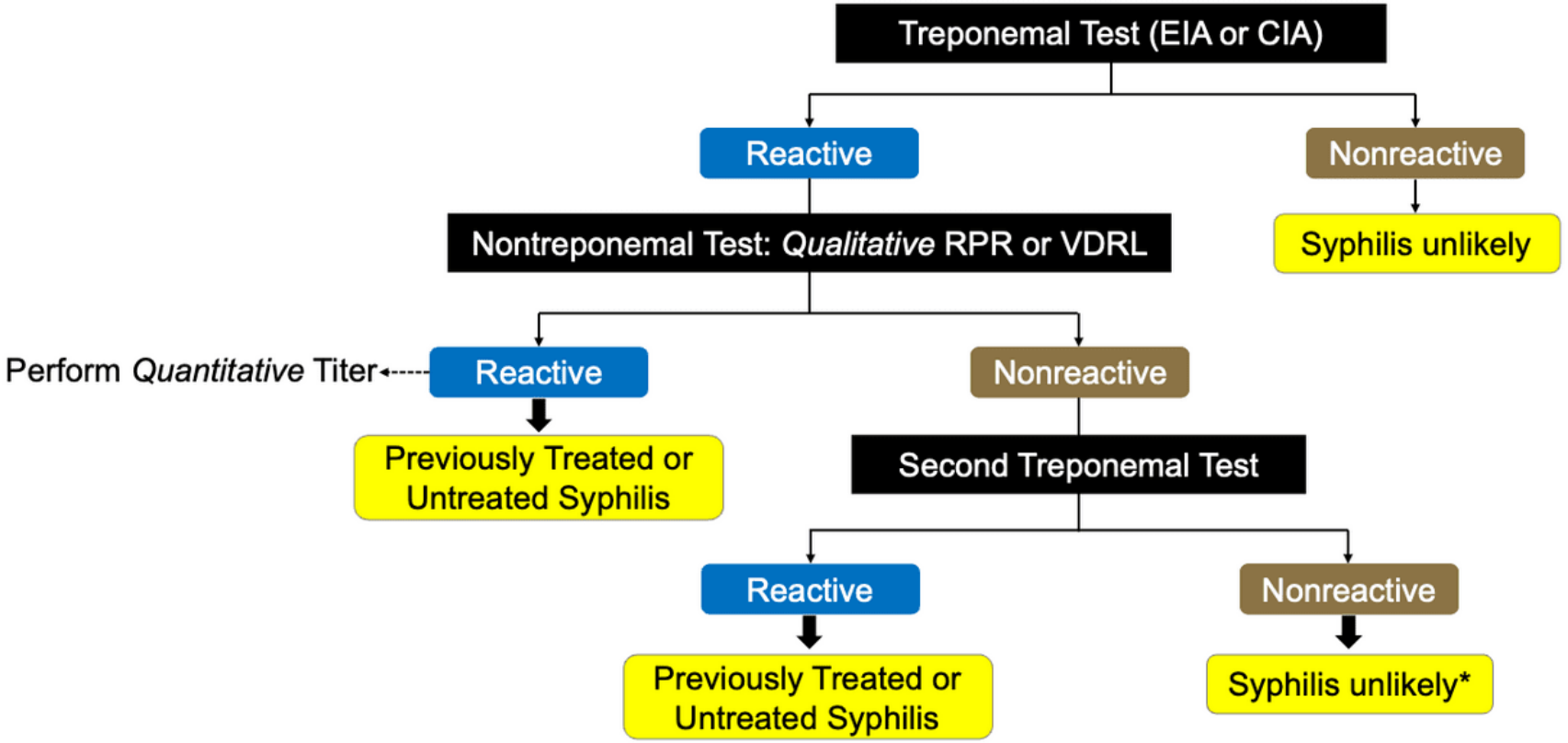
Syphilis: Diagnosis

- *Treponema pallidum* is difficult to culture, so testing relies on indirect measures:

- **Treponemal-specific tests= qualitative**
  - –EIA, TPPA, FTA-ABS among others

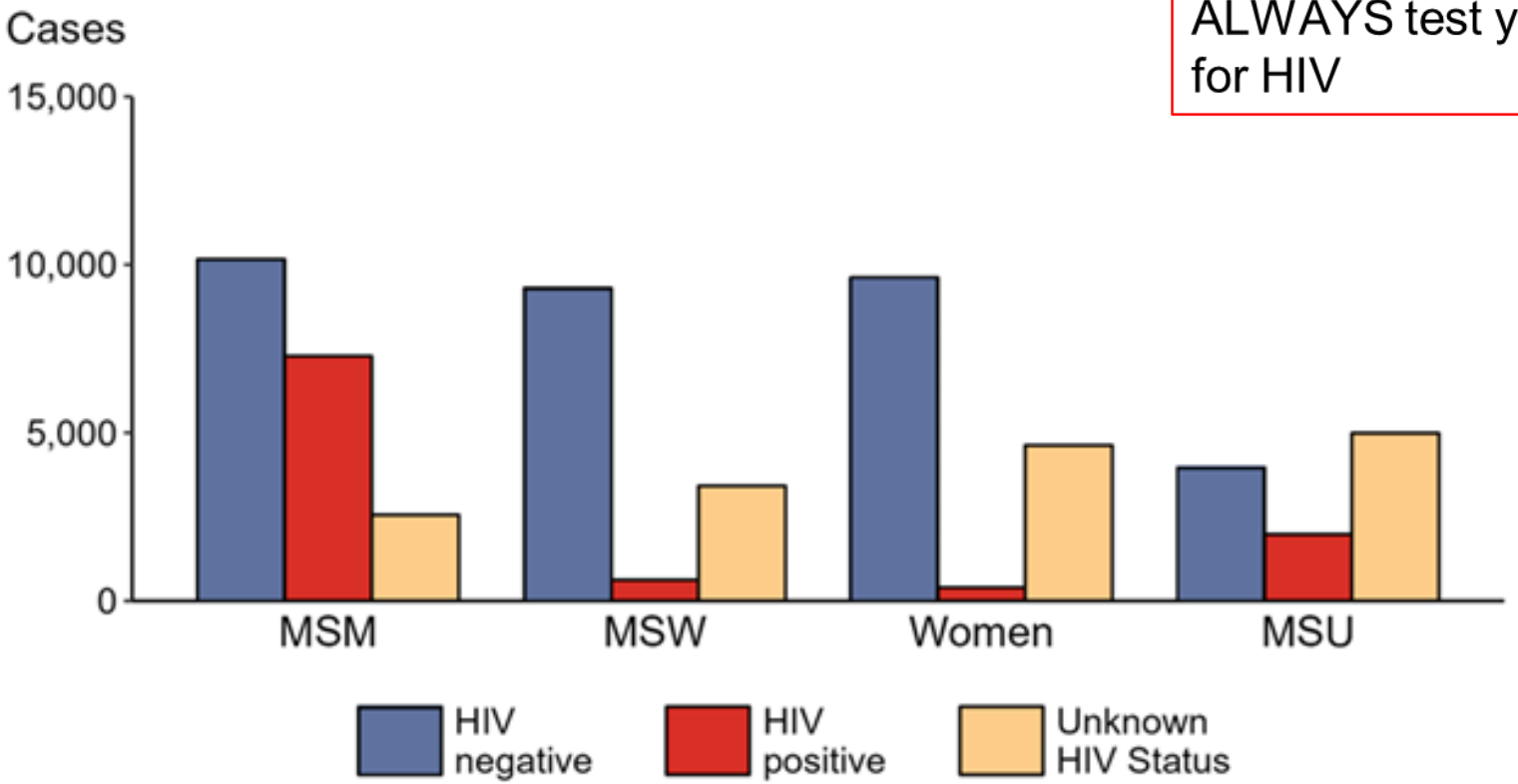
- **Non-treponemal tests- quantitative**
  - –RPR (rapid plasma reagin) or VDRL (venereal disease research laboratory)

Syphilis: Diagnosis



Primary and Secondary Syphilis — Reported Cases by Sex, Sex of Sex Partners, and HIV Status, United States, 2022

ALWAYS test your patient for HIV



- **ACRONYMS:** MSM = Men who have sex with men; MSW = Men who have sex with women only; MSU = Men with unknown sex of sex partners

Syphilis: Treatment

- Primary/ Secondary/ Early Latent
  - **Penicillin** G 2.4 million units IM x 1 dose
  - Pen allergic: Doxycycline 100 mg bid x 14 days
- Tertiary/ Late Latent
  - Penicillin G 2.4 million units IM weekly x 3 doses
  - Pen allergic: Doxycycline 100 mg bid x 28 days
- Neuro/ Ocular
  - **aqueous penicillin G** 4 million units every 4 hours x 14 days

Syphilis: Treatment

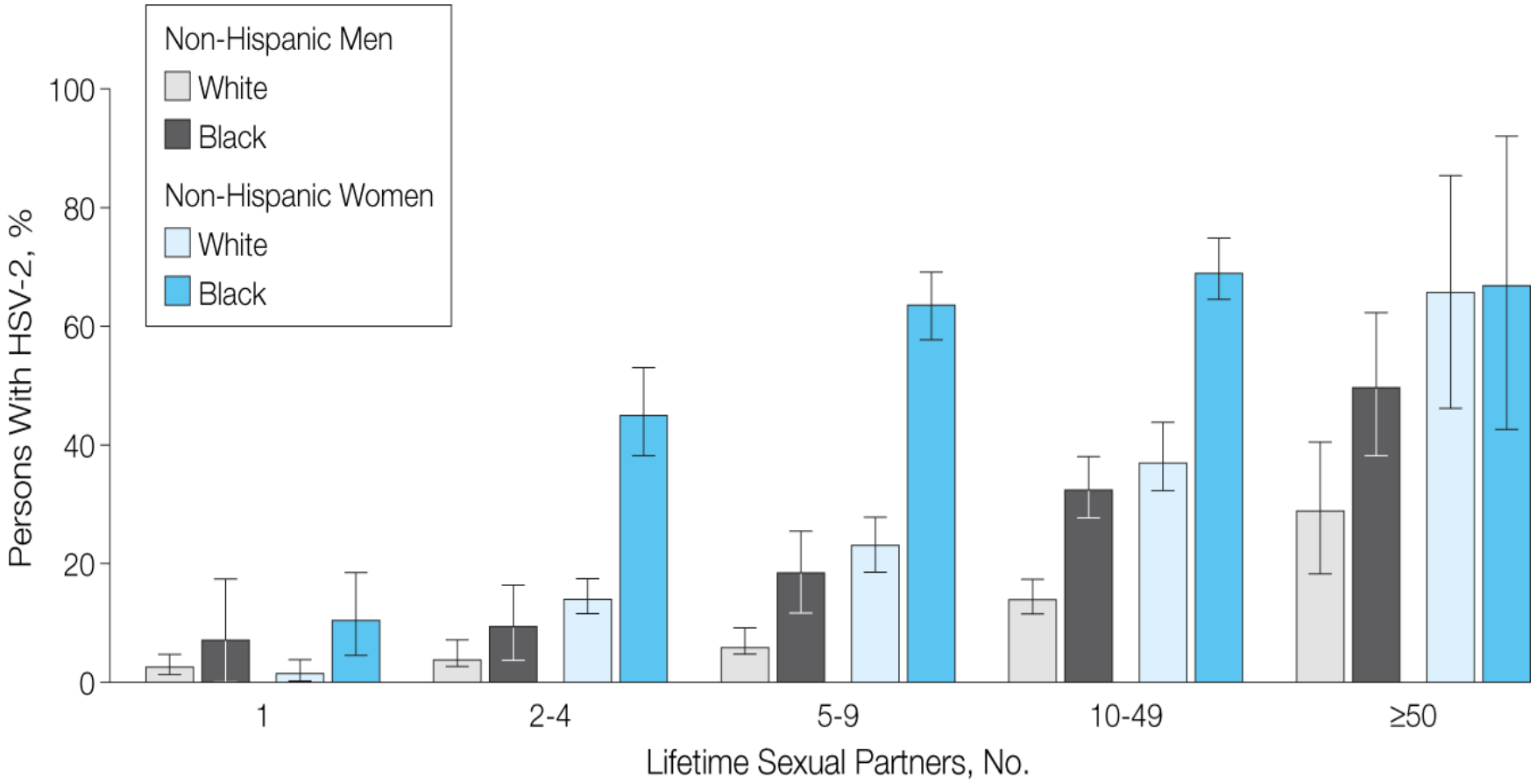
- **Jarisch- Herxheimer Reaction**
  - –Acute syndrome that occurs as a result of inflammation from dying spirochetes
  - –Symptoms include fever, myalgias, headaches
  - –Self-limiting
  - –All patients should be counseled to monitor for this reaction prior to initiating therapy

Herpes Simplex Virus

HSV: Virology

- HSV is virus that infects the skin and mucosa
  - HSV-1 has typically been associated with oral mucosa whereas HSV-2 has been associated with genital mucosa
    - However, there has been a rise in the prevalence of HSV-1 genital lesions
    - **HSV-2 increases risk of HIV acquisition**
  - Seroprevalence of both viruses **increases with age**
    - HSV-1 >85% adults have been exposed
    - HSV-2 >25% of adults have been exposed

HSV-2 Prevalence in US based on Lifetime Sexual Partners



HSV: Presentation

- Clinical findings:
  - –Small cluster or group of vesicular lesions
    - •Vermillion border of lip, penile shaft, labia, perianal skin, buttocks
    - •Lesions typically crust and heal within 1 week
    - •**Painful**
  - –Neuralgia (burning, stinging)
  - – Local lymphadenopathy

HSV-Genital herpes





HSV-Perianal herpes in immunocompromised host



HSV: Presentation

- Clinical findings:
  - **Primary infection** occurs in a person without pre-existing antibodies upon exposure to the virus
  - **Recurrent infection** occurs as an “outbreak” in the setting of know seropositivity
  - Can be symptomatic or asymptomatic

HSV: Presentation

- Other clinical manifestations include:
  - Neurologic: meningitis, encephalitis
  - Ocular: keratitis, blepharitis, conjunctivitis
  - GI: esophagitis, hepatitis
  - Skin: erythema multiforme

HSV-Herpetic whitlow





HSV: Diagnosis

- Clinical
- PCR of fluid from unroofed vesicle
- Viral culture, sensitivity <50%
- Histology

Do not screen asymptomatic patients unless undergoing immunosuppression!

HSV: Treatment

- Initiating antivirals within 72 hours of an outbreak can decrease the severity and length of illness
  - Acyclovir 400 mg 3x/day
  - **Valacyclovir 1g daily or 2x/day**
  - Famciclovir 250 mg 3x/day
  - Duration: 7-10 days

Other Genital Ulcer Diseases

- Lymphogranuloma venereum (LGV)
- Granuloma inguinale
- Chancroid

Other Genital Ulcer Diseases

- Lymphogranuloma venereum (LGV)
  - –*Chlamydia trachomatis* serotypes L1-L3
  - –Single **painless** ulcer with PAINFUL LAD
  - –Proctitis (can be mistaken for IBD)
  - –NAAT testing→ multiplex PCR
  - –Doxycycline daily or azithromycin x3 weeks

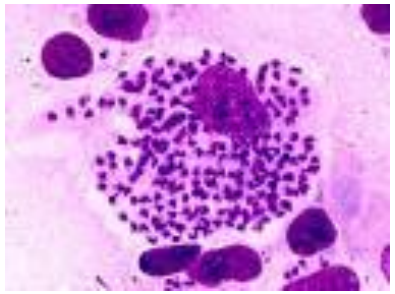


Source: Kenneth J. Ryan
Sherris & Ryan's Medical Microbiology, 8e
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Other Genital Ulcer Diseases

- Granuloma inguinale
  - –Donovanosis caused by *Klebsiella granulomatis*
  - –**Painless**, destructive, serpiginous lesion
  - –Beefy red with white border, highly vascular with bleeding on biopsy
  - – Tissue biopsy
    - Donovan bodies: Wright-Giemsa stain
  - –Doxycycline or azithromycin until lesions resolve





Other Genital Ulcer Diseases

- Chancroid
  - –*Haemophilus ducreyi*
    - Outbreaks in MSM and HIV +
  - –**Painful**, indurated, ragged ulcer
  - –"kissing lesions" on thigh
  - –Painful LAD
  - –Diagnosis based on culture
  - –Azithromycin 1g or ceftriaxone 250 mg IM x1



Human Papilloma Virus
also an STI
- refer to prior block for review

References

- @CDCSTD
- 2021 Sexually Transmitted Diseases Guidelines www.cdc.gov
- 2024 Sexually Transmitted Disease Surveillance www.cdc.gov
- National STI Curriculum https://www.std.uw.edu/
- Clinical Infectious Disease (2nd ed)
- ScholarRx

Questions?

Vaginitis

Danielle Cullen, MD

October 29, 2026

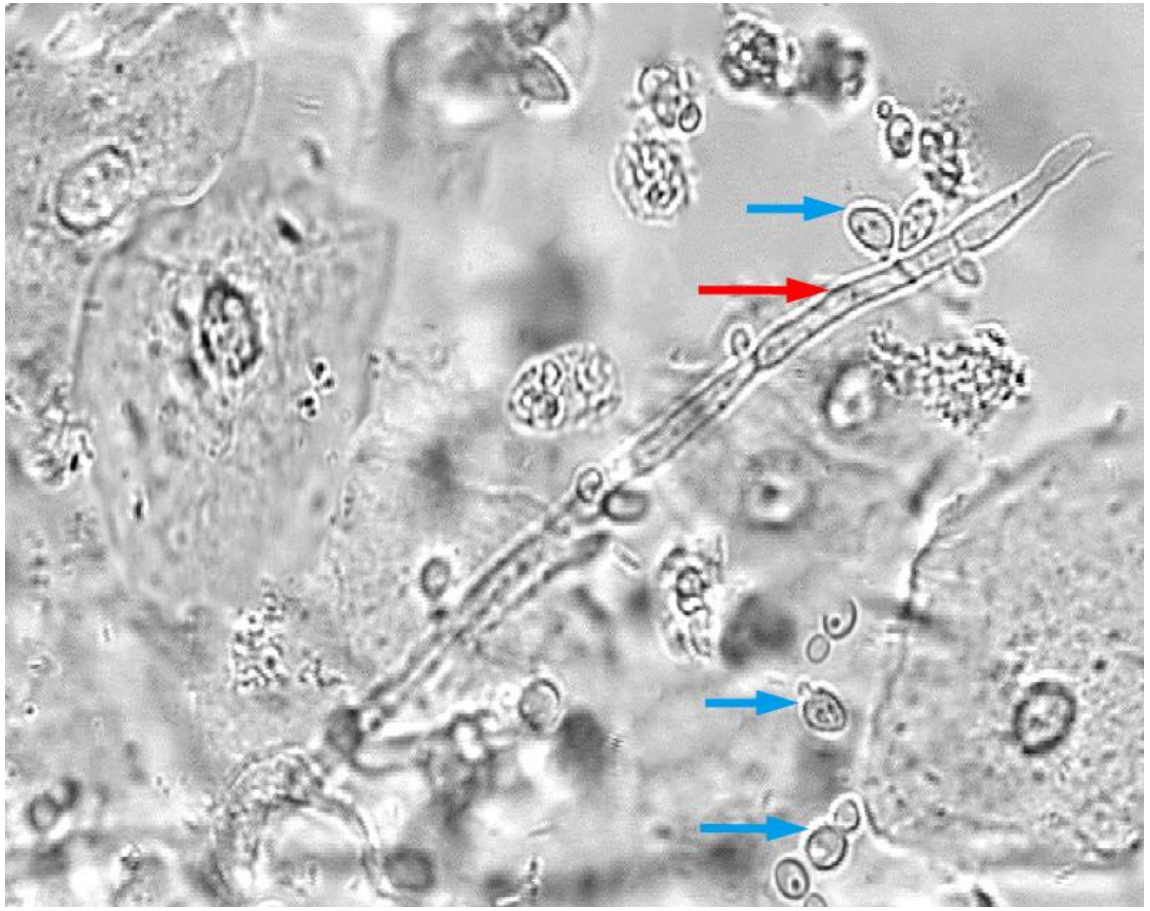
Learning Objectives

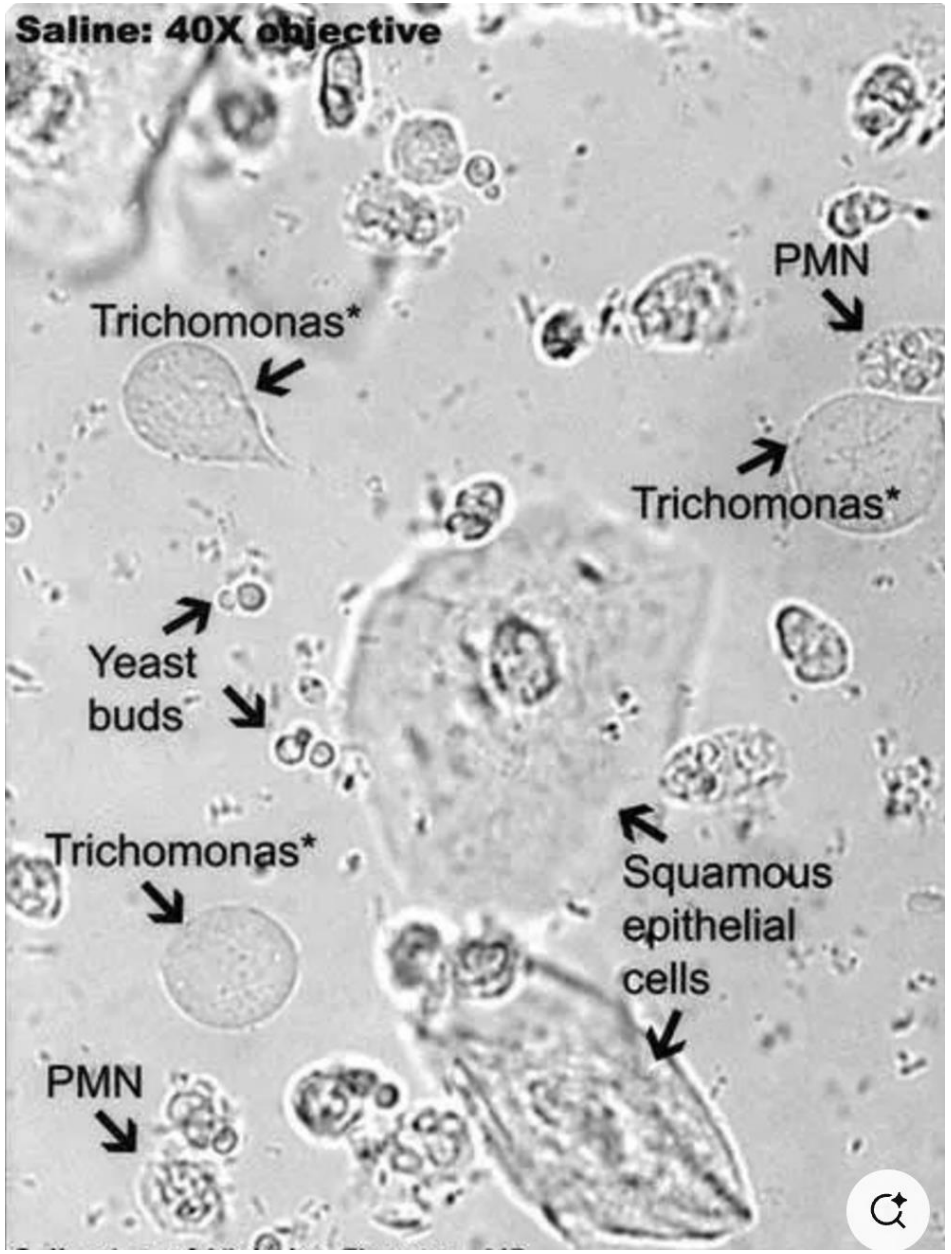
- After attending this session, attendees should be able to
  1. Develop a differential diagnosis for a patient with a chief complaint of "vaginal discharge"
  2. Describe the microbiology and diagnostic findings (including microscopy) of yeast vaginitis, bacterial vaginosis, and trichomoniasis
  3. Differentiate these infections by their clinical presentation and physical exam findings
  4. Describe the key epidemiologic features and standard treatments for each

Chief Complaint: Vaginal Discharge

- Differential
  - STIs: GC, CT, **trichomonas**, HSV (especially primary)
  - **Bacterial vaginosis (BV)**
  - **Vaginal candidiasis ("yeast infection")**
  - Retained foreign body
  - Desquamative inflammatory vaginitis (DIV)
  - Structural lesion
    - Prolapsing polyp or fibroid
    - Necrosing intracavitary fibroid
    - Cervical cancer

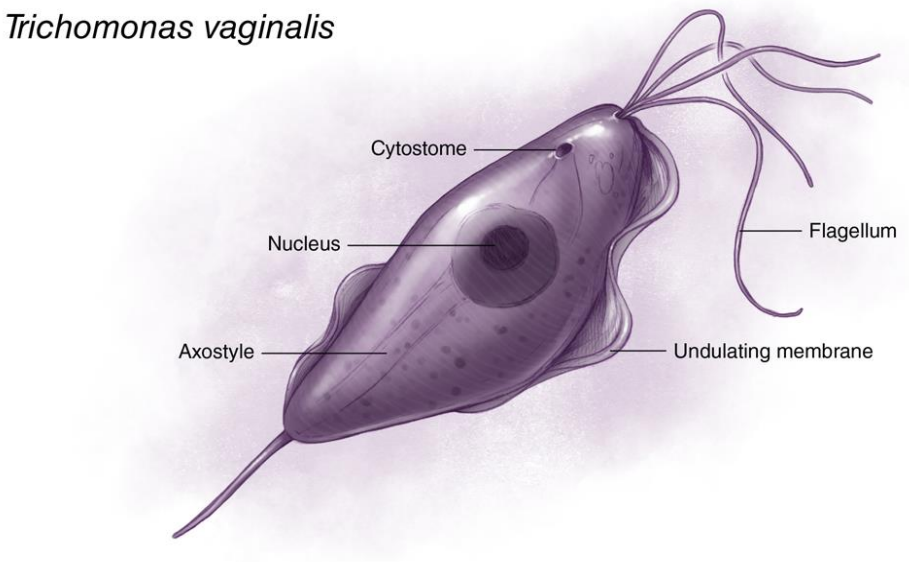
Sizes of cells/microbes





Trichomoniasis:
Protozoan *Trichomonas vaginalis*

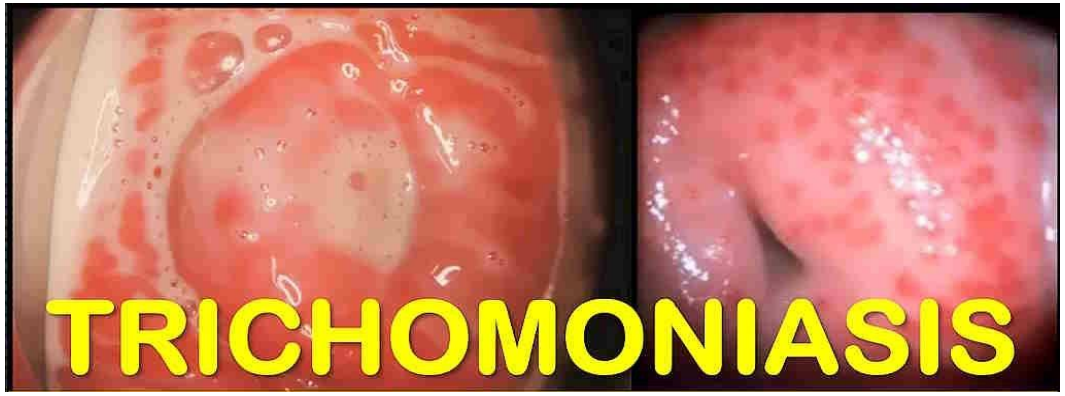
- Single-celled flagellated anaerobic protozoan parasite
- Only known protozoan parasite that infects the genital tract
- Four anterior flagella and one posterior flagellum embedded in the "undulating membrane" --> jerky movement under the microscope
- Unlike other protozoa (specifically flagellates), *T. vaginalis* has no cyst form, and its trophozoites cannot survive outside the host
  - Thus *T. vaginalis* is only transmitted directly from person to person via sexual contact
- If asymptomatic, may persist for months to years in the vagina



The **trophozoite** is the active, growing, feeding stage in the protozoan life cycle

The cyst form is dormant and protected, but T. vaginalis has no cyst form

Trichomoniasis: Symptoms and Physical Exam



- Symptoms
  - Yellow-green vaginal discharge
  - Vaginal (+/- vulvar) pruritus
  - Asymptomatic
- Physical examination
  - **Frothy** yellow-green discharge
  - "Strawberry cervix"- cervical punctate hemorrhages, only in 2-5% of cases but is strongly suggestive of TV when present
  - Chronic infection may be associated with minimal vaginal discharge, mild pruritus, and/or dyspareunia

Trichomoniasis: Epidemiology

- Almost always sexually transmitted (no dormant cyst form)
- Infection associated with increased risk of acquiring HIV, as well as PID, infertility and cervical cancer
  - Up to 53% of women with HIV have *T. vaginalis* infection
- ~ 3.7 million people have trichomoniasis in the US
  - The **prevalence** of *T. vaginalis* infection among people with vaginas of reproductive age in the US is ~2%
- 1.1 million new cases occurring each year

- Men who have sex with men **appear to be** at low risk of acquiring trichomoniasis (**not screened**)
- Transmission of *T. vaginalis* betweer
- CDC screening recommendations:

Trichomonas

Women	• Consider screening for women receiving care in high-prevalence settings (e.g., STI clinics and correctional facilities) and for asymptomatic women at high risk for infection (e.g., women with multiple sex partners, transactional sex, drug misuse, or a history of STI or incarceration)[2]
Persons with HIV	• Recommended for sexually active women at entry to care and at least annually thereafter[2,6]

Trichomoniasis: Testing

- Microscopy: motile trophozoite and many PMNs
  - Only motile for 10-20 min after collection!
- Vaginal pH elevated (5-6)
- NAAT
- Culture is less accurate (sensitivity 43-95%)



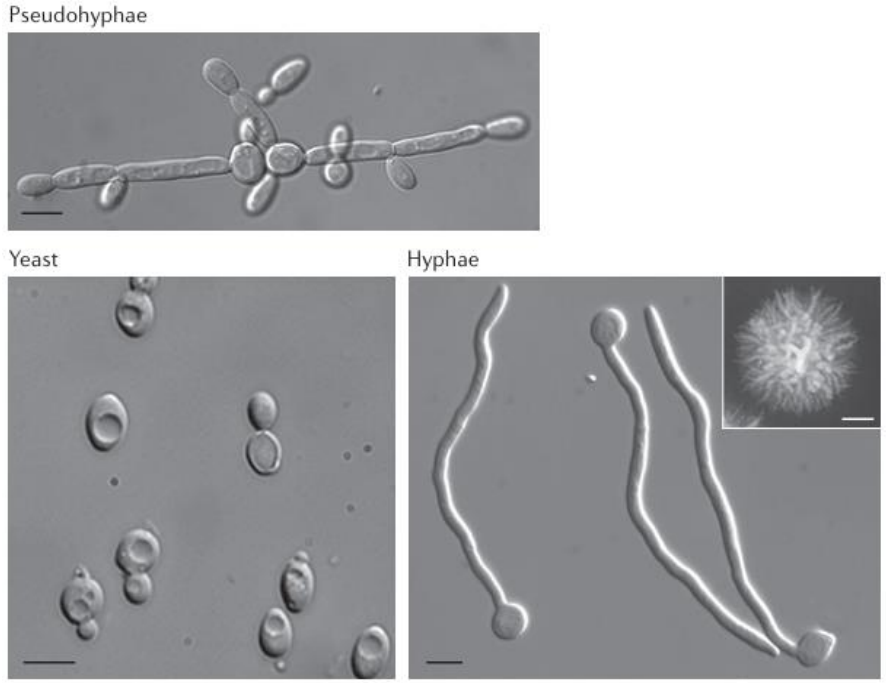
Trichomoniasis: Treatment

- Metronidazole 500mg BID x 7 days (2g once dosing is less effective)
  - Antibacterial (anaerobes) and antiprotozoal medication

- Alternative: tinidazole 2g once daily for 7 days
- Partner treatment needed
- Test of cure is recommended by the CDC, about 3 months after treatment
  - Do not retest until **at least** 3 weeks after treatment completion

Yeast Vaginitis: microbiology

- Vulvovaginal candidiasis
  - ~90% is Candida albicans
    - Hyphae, pseudohyphae, and budding forms
  - The rest are "non-albicans" (C. glabrata, C. krusei, C. tropicalis, C. parapsilosis)



- Hyphae are long, tube-like and without constrictions
- Pseudohyphae have constrictions at sites of septation and are wider
- Budding yeast are unicellular

Yeast vaginitis: Symptoms and Physical Exam

- Clumpy/cottage cheese like discharge
- Vulvar and/or vaginal itching
- External rash and irritation- especially in poorly controlled diabetes and immune suppressed individuals



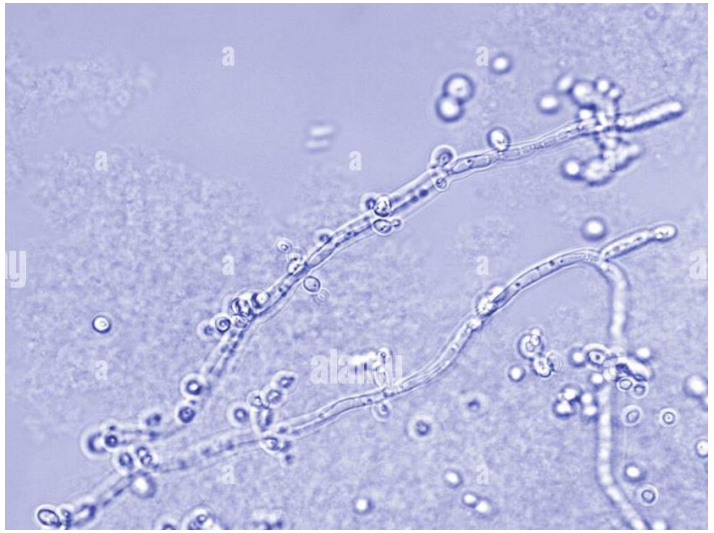


Yeast vaginitis: Epidemiology

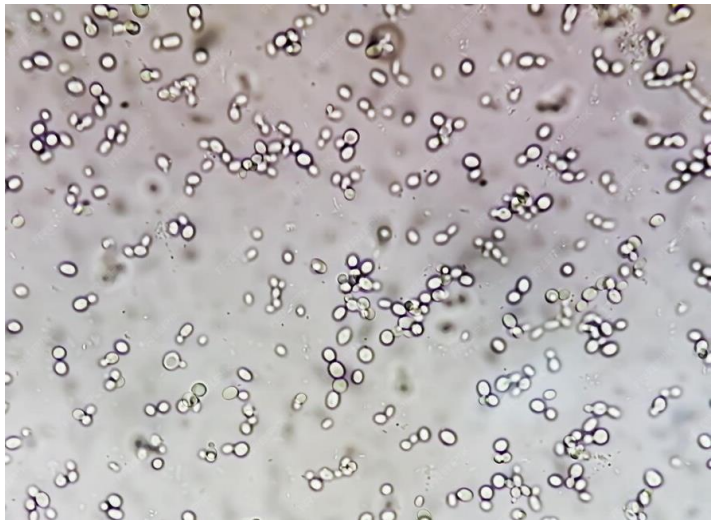
- Candida are part of the normal vaginal flora in up to ¼ of people
- Risk factors for symptomatic candida include recent antibiotic use, immune suppression, and diabetes
- Candida is the 2nd most common cause of abnormal vaginal discharge, about 1/3 of cases
- Difficult to study epidemiology due to easily available OTC treatments
  - Surveys show that up to 55% of people with vaginas have been dx with yeast at some point

Yeast Vaginitis: Testing

- Microscopy: Hyphae, pseudohyphae, budding yeast, many WBC (PMNs)
- Vaginal pH is normal (~4.5)



- NAAT
- Yeast culture
  - Used for recurrent or persistent infections
  - Can differentiate species



Yeast vaginitis: Treatment

- Uncomplicated: infrequent, non-pregnant, immunocompetent
  - Oral Fluconazole 150mg once
  - Topical Azoles- prefer 7d, can use 3d
  - Boric Acid- **vaginal** suppository x 7 days

- No need to treat if asymptomatic (for example, noted on pap result)

DON'T MEMORIZE THIS CHART!

Treatment of uncomplicated vulvovaginal candidiasis

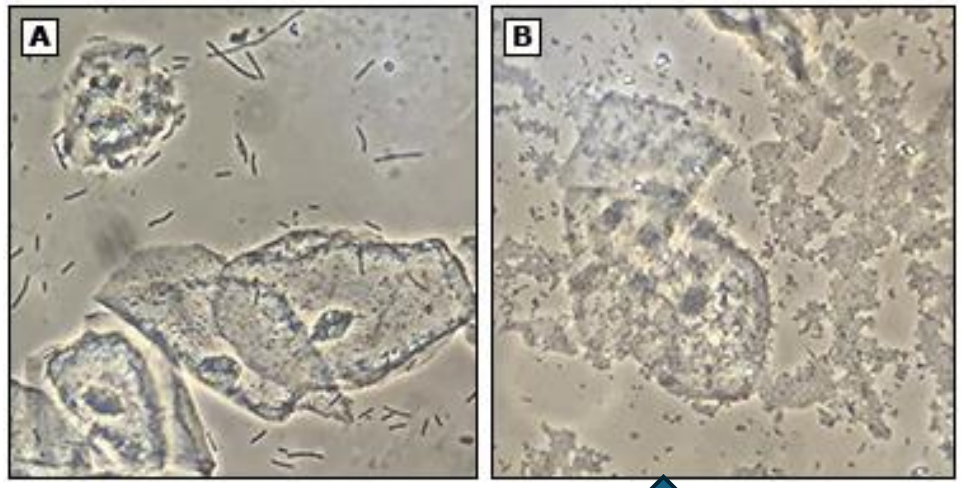
Drug and trade name(s)	Requires a prescription in US	Preparation	Intravaginal* dose for adult
Fluconazole oral administration¶			
Diflucan△	Yes	150 mg oral tablet	Single dose by mouth; patients with greater vulvovaginal involvement may require an additional dose of 150 mg given 72 hours after initial dose
Clotrimazole			
Gyne-Lotrimin△	No	1% vaginal cream	1 applicatorful (approximately 5 g) once daily at bedtime for seven days
Gyne-Lotrimin 3△	No	2% vaginal cream	1 applicatorful (approximately 5 g) once daily at bedtime for three days
Gyne-Lotrimin◇	Not applicable (not available in US)	100 mg vaginal tablet◇ 200 mg vaginal tablet◇ 500 mg vaginal tablet◇	Insert one 100 mg vaginal tablet once daily for seven days; or One 200 mg tablet once daily for three days; or One 500 mg tablet as a single dose
Canesten 1 day internal cream◇	Not applicable (not available in US)	10% vaginal cream (combination kit may include 1% clotrimazole cream for external use)◇	1 prefilled applicatorful (approximately 5 g) at bedtime as a single dose
Miconazole			
Monistat 7△	No	2% vaginal cream (combination kit may include 2% miconazole cream for external use)	1 applicatorful (approximately 5 g) once daily at bedtime for seven days
Monistat 3△	No	4% vaginal cream	1 applicatorful (approximately 5 g) once daily at bedtime for three days
Monistat 7△	No	100 mg vaginal suppository	1 suppository once daily at bedtime for seven days
Monistat 3△, Vagistat-3△	No (combination kit) Yes (generic suppository)	200 mg vaginal suppository (combination kit may include 2% miconazole cream for external use)	1 suppository once daily at bedtime for three days
Monistat 1△	No	1200 mg vaginal suppository (combination kit may include 2% miconazole cream for external use)	1 suppository for one day as a single dose
Nystatin§			
Nystatin vaginal◇ (former US brand Mycostatin)	Commercial vaginal tablet not available in US; vaginal suppositories can be prepared by a licensed compounding pharmacy (prescription required)	100,000 unit vaginal tablet or suppository	Insert 1 vaginal tablet or suppository once daily for 7 days
Terconazole¥			
Generic only (former US brands Terazole 7, Zazole)△	Yes	0.4% vaginal cream	1 applicatorful (approximately 5 g) once daily at bedtime for seven days
Generic only (former US brands Terazole 3, Zazole)△	Yes	0.8% vaginal cream	1 applicatorful (approximately 5 g) once daily at bedtime for three days
Generic only (former US brands Terazole 3, Zazole)△	Yes	80 mg vaginal suppository	1 suppository once daily at bedtime for three days
Tioconazole			
Vagistat-1, Monistat 1-Day△	No	6.5% vaginal ointment	1 applicatorful (approximately 5 g) at bedtime as a single dose
Butoconazole			
Gynazole-1	Yes	2% vaginal cream	1 applicatorful (approximately 5 g) as a single dose
Ibrexafungerp‡			
Brexafemme	Yes	150 mg oral tablet	Two 150 mg tablets (ie, 300 mg) by mouth twice a day for one day (4 tablets total). Limit use to **non**-pregnant patients who prefer oral dosing but cannot use oral fluconazole (eg, due to intolerance, allergy, or *Candida* infection resistant to fluconazole). Dose reduction required if taken with strong CYP3A inhibitors.
Boric acid			
AZO Boric acid, others*	No	600 mg vaginal suppository	1 suppository once daily at bedtime for 7 days. Limit use to patients who are allergic to fluconazole and do not have access to other topical agents (ie, azoles and nystatin)†. **Warning: boric acid may cause death if taken orally.**

Yeast Vaginitis: Treatment

- Complicated
  - Pregnancy: 7 day topical azoles (clotrimazole and miconazole first line), may need 10-14 day treatment
  - Immunocompromise: Fluconazole oral 150mg Q72h for 2-3 doses, or 14 day course of topical azole
  - Non-albicans: Vaginal boric acid 600mg vaginally for 14-21 nights

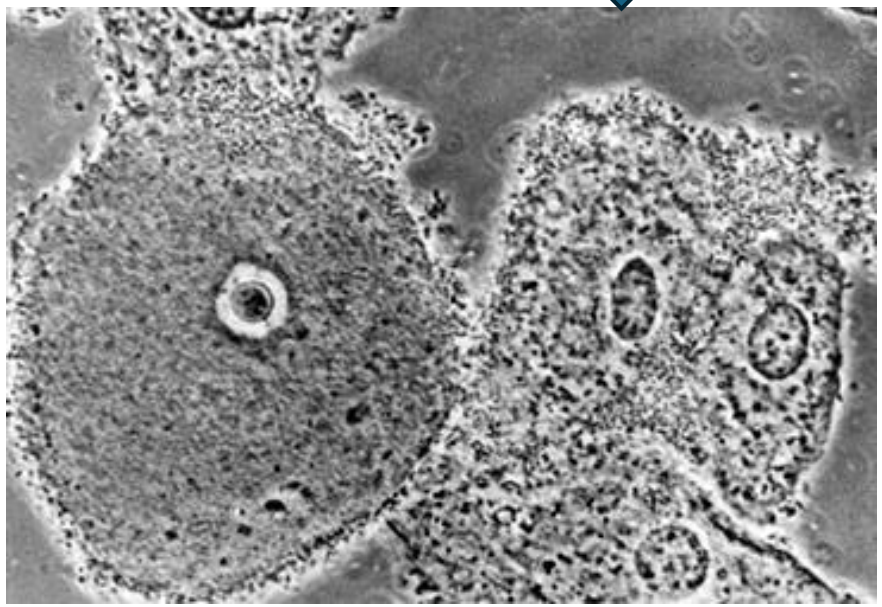
BV: Microbiology

- Shift from Lactobacillus predominance to bacterial diversity
  - Facultative anaerobes, especially gram-negative rods
  - Common species: *Gardnerella vaginalis*, *Mycoplasma hominis*, *Ureaplasma urealyticum*, *Mobiluncus, Megaspera, Prevotella* species, *Bacteroides* species, *Atopobium vaginae* etc
- Anaerobes make proteolytic carboxylase enzymes-->break down vaginal peptides into volatile amines (malodorous & cause increased discharge and squamous cell exfoliation)
- *Gardnerella* adheres to the squamous cells and creates a biofilm, which becomes a scaffold for other bacteria to adhere to



NORMAL

BV



BV: Microbiology

- Why does the bacterial shift happen?

- Wiping out/killing of Lactobacillus
  - Douching
  - Cleaning inside the vagina with soap or other products (vaginal cleansers)
  - Prolonged submerging in warm/hot water
  - Certain sex partners? Exposure to BV associated pathogens

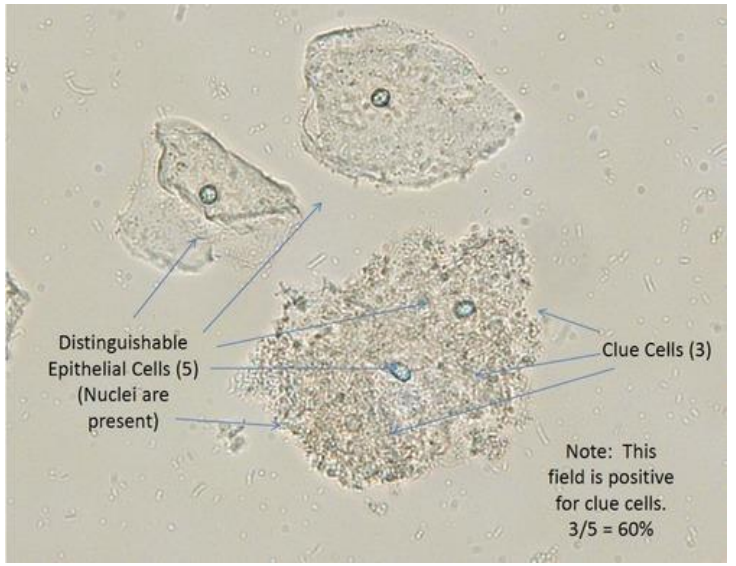
BV: Symptoms and Physical Exam

- Not inflammatory, thus "vagin**osis**"
- Symptoms: vaginal discharge, odor, +/- mild irritation
- Speculum exam: Thin, malodorous off-white discharge

BV: epidemiology

- Makes up 40-50% of abnormal vaginal discharge visits
- Prevalence is 29-50% in the general population (self-swab, symptomatic and asymptomatic)
  - Should they be treated if asymptomatic/found incidentally??
- Increases risk of preterm birth, post-birth infection, infections after vaginal surgery, acquisition and transmission of HIV and other STIs, development of PID, and persistence of HPV

BV: Testing



- Clinical Diagnosis: Amsel Criteria (need 3 of 4)
  1. Thin homogenous discharge
  2. Microscopy: Clue cells (at least 20%)
  3. PH >4.5 (due to increased production of volatile amines and decreased production of lactic acid and hydrogen peroxide)
  4. Fishy odor with 10% potassium hydroxide (+ whiff-amine test)

- Laboratory testing
  1. NAAT (various FDA approved and lab-developed tests, testing for bacteria common in BV)
  2. Gram stain (gold standard, but mostly research now, minimal clinical utility)



BV: Treatment



- Metronidazole 500mg BID x 7 days
- Metronidazole 0.75% gel vaginally QHS 5 days

- Partner treatment??
  - Newer data suggest that it significantly decreases recurrence

- Vulvovaginal care adaptations
  - Use unscented soap
  - Avoid products inside the vagina (no douching!!)
  - Avoid scented products (tampons, pads, deodorizers)
  - Avoid tight pants or prolonged wet clothing
  - Use cotton underwear, avoid thongs, sleep without underwear

Vaginitis/vaginosis Overview

Infection	PH	Microscopy	Symptoms	Exam
Candidal vaginitis (yeast)	*Normal (3.8-4.5)	**KOH or wet mount- budding forms and/or hyphae, increased PMNs	Clumpy discharge, itching, irritation, vulvar/groin rash	Thick, clumpy, adherent, white/yellow
Bacterial vagin**osis** (BV)	High (>4.5)	Clue cells Few to no lactobacilli	Discharge, fishy/garbage-like odor, irritation	Thin, grey/off-white, malodorous, homogenous
Trichomonas vaginalis	High (5-6)	Motile trichomonads, many PMNs	Thin discharge, burning, itching or irritation	**Frothy,** thin, malodorous, yellow-green, strawberry cervix (2-5% of cases)

- * normal vaginal pH is ACIDIC! Due to lactic acid and hydrogen peroxide production by lactobacilli
- ** KOH- 10% potassium hydroxide

Questions??

Great resource!

https://www.std.uw.edu/go/comprehensive-study/vaginitis/core-concept/all